

Covolume-upwind finite volume approximations for linear elliptic partial differential equations

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ABSTRACT

In this paper, we study covolume-upwind finite volume methods on rectangular meshes for solving linear elliptic partial differential equations with mixed boundary conditions. To avoid non-physical numerical oscillations for convection-dominated problems, nonstandard control volumes (covolumes) are generated based on local Peclet's numbers and the upwind principle for finite volume approximations. Two types of discretization schemes with mass lumping are developed with use of bilinear or biquadratic basis functions as the trial space respectively. Some stability analyses of the schemes are presented for the model problem with constant coefficients. Various examples are also carried out to numerically demonstrate stability and optimal convergence of the proposed methods.

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1. Introduction

The second-order linear elliptic partial differential equations, or say the convection–diffusion–reaction equations, are of intensive applications in various fields of science and engineering, e.g., fluid mechanics, groundwater prediction, environmental protection, and so on. The finite element method (FEM) and the finite volume method (FVM) have been widely used to numerically solve these problems in practical simulations. An advantage of finite volume method is that one can convert integrations over local control volumes (also called covolumes) to integrations over their boundaries based on the Green's theorem, finite difference or quadrature rules then can be used to approximate the resulting terms in local elements. The discretization schemes derived from finite volume methods usually satisfy local mass conservation, which makes the finite volume method an important and effective tool to numerically simulate various physical models [5,6,26,3,21,8,25].

In many of convection–diffusion type problems arising from real applications, the convection coefficients are often much larger than the diffusion ones, and consequently numerical accuracies of the approximate solutions found by central finite volume discretizations (CFVM) using standard control volumes are often compromised by nonphysical numerical oscillations [14,9]. In order to overcome this difficulty upwind finite volume methods (UFVM) were proposed for solution of convection-dominated problems and had been quite successful in flow and heat simulations [11,24,28,7,4,13]. Although being numerically stable for solving this type of problems, the standard upwind approximations suffer from bad accuracy mostly due to their first-order local truncation errors. Lots of modifications on the standard upwind finite volume scheme

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including adaptive upwind schemes have been introduced in order to regain the second or higher-order accuracies, see [1,10,12,23,20]. In [14], Lazarov et al. studied finite volume methods for convection–diffusion problems over rectangular meshes, L^2 – and H^1 –errors are derived for cell-centered scheme and various upwind schemes. For similar methods on triangular meshes, we refer to [18,22] for detailed discussions. With use of these modified upwind schemes, people often can partially recover high-order accuracy, but many of them still either do not satisfy the maximum principle and/or the stability condition, thus again produce new numerical oscillations in the discrete approximation, or still introduce numerical dispersions for strongly convection-dominated cases. In [16], an optimal weighted upwinding covolume method (OWUCM) on rectangular grids was proposed for solving second-order linear elliptic problems with pure Dirichlet boundary conditions in the two-dimensional space. Instead of using the standard dual central grid as the control volumes, some nonstandard dual grids determined by the local *Peclet's number* is used to define the control volumes, which will also be a starting point of the proposed work in this paper. However, in order to force the resulting linear system to be a M -matrix (thus the OWUCM scheme satisfies the discrete maximum principle), local integrations used for the purpose of mass lumping in the OWUCM scheme is only first-order accurate when the problem is a convection-dominated one, and thus it still could lose the second-order accuracy measured in the L^2 norm like the standard upwind schemes.

In this paper, we propose and study bilinear and biquadratic *covolume-upwind finite volume methods* (CUFVM) on rectangular grids for solving linear elliptic problems with mixed boundary conditions in two dimensional space. For the method based on bilinear basis functions, we use a similar way as suggested in [16] to construct nonstandard and upwind dual grids as the control volumes, whose vertices can be automatically computed using the local Peclet's numbers and may vary a lot across the whole region in the relative positions to their primary grid points. On the other hand, our discretization scheme basically uses the middle point quadrature for mass lumping and thus the local truncation errors are of second order. The idea is then naturally extended to the case using biquadratic basis functions [17] to obtain a higher-order convergent scheme. Through some analysis and various numerical experiments, we demonstrate that the proposed methods are not only numerically stable but also have optimal convergence, even for strongly convection-dominated problems. Here the optimal convergence means that for the bilinear CUFVM, the convergence is of second order with respect to the grid size when the errors are measured under the L^2 norm and is of first order under the H^1 norm, and for the biquadratic CUFVM, the convergence is of third order under the L^2 norm and of second order under the H^1 norm. We also would like to note that the proposed CUFVMs can be easily extended to three dimensions.

This paper is organized as follows. The model problem is first introduced with some basic properties in Section 1.1. Then in Section 2, we present and analyze the bilinear covolume-upwind finite volume method, including construction of the upwind control volumes, numerical discretizations with local mass lumping, and some stability analyses of the scheme for problems with constant coefficients. In Section 3, the proposed method is further generalized to the case using biquadratic basis functions including derivation of upwind control volumes and corresponding numerical approximations for the biquadratic finite volume scheme. Various numerical experiments are carried out in Section 4 to demonstrate stability and optimal convergence of the proposed methods, followed by conclusions given in Section 5.

1.1. The model problem

Let $\Omega \in \mathbb{R}^d$ be a bounded domain with piecewise polygonal boundary $\partial\Omega$. We consider the following second-order convection–diffusion–reaction equation with mixed boundary conditions:

$$\begin{cases} \nabla \cdot (-a\nabla u + \vec{b}u) + ru = f, & \text{in } \Omega, \\ u = g, & \text{on } \Gamma_D, \\ (-a\nabla u + \vec{b}u) \cdot \vec{n} = h_{in}, & \text{on } \Gamma_N^{in}, \\ -a\nabla u \cdot \vec{n} = h_{out}, & \text{on } \Gamma_N^{out}, \end{cases} \quad (1)$$

where $a(\mathbf{x}) > \tilde{a} > 0 \in H^1(\Omega)$ is the diffusion rate, $\vec{b}(\mathbf{x})$ represents the transport velocity such that $\nabla \cdot \vec{b}(\mathbf{x}) \in L^\infty(\Omega)$, $\vec{n}$ is the unit outward normal to $\partial\Omega$, $r(\mathbf{x}) \geq 0 \in L^\infty(\Omega)$ is the reaction rate, and $f(\mathbf{x})$ denotes the given source function. This problem is a prototype for flow and transport in porous media where $u(\mathbf{x})$ often represents the pressure head in an aquifer or the concentration of a chemical in ground water. The domain boundary $\partial\Omega$ is the union of the Dirichlet boundary $\Gamma_D \neq \emptyset$, the in-flow boundary $\Gamma_N^{in} = \{\mathbf{x} \in \partial\Omega | \vec{b}u \cdot \vec{n} \leq 0\}$ and the out-flow boundary $\Gamma_N^{out} = \{\mathbf{x} \in \partial\Omega | \vec{b}u \cdot \vec{n} \geq 0\}$. Let us set $\Gamma_N = \Gamma_N^{in} \cup \Gamma_N^{out}$.

It is also assumed that

$$\frac{\nabla \cdot \vec{b}}{2} + r \geq 0. \quad (2)$$

To investigate well-posedness of the problem (1), let us only consider the case with homogeneous boundary conditions $g = 0$ and $h_{out} = 0$. Define Sobolev space $H_D^1(\Omega) = \{v \in H^1(\Omega) | v = 0 \text{ on } \Gamma_D\}$. The corresponding weak form for the problem (1) is then to find $u \in H_D^1(\Omega)$ such that

$$A(u, v) = L(v), \quad \forall v \in H_D^1(\Omega), \quad (3)$$

where

$$A(u, v) = \int_{\Omega} (a \nabla u - \vec{b}u) \cdot \nabla v \, d\mathbf{x} + \int_{\Omega} ruv \, d\mathbf{x} + \int_{\Gamma_N^{out}} (\vec{b} \cdot \vec{n})uv \, ds, \quad (4)$$

$$L(v) = \int_{\Omega} f v \, d\mathbf{x} - \int_{\Gamma_N^{in}} h_{in} v \, ds. \quad (5)$$

It is also known that the bilinear form $A(u, v)$ has the following properties provided that the assumption (2) is satisfied:

$$A(u, u) \geq c_1 \|u\|_{H^1}^2, \quad \forall u \in H_D^1(\Omega), \quad (6)$$

$$A(u, v) \leq c_2 \|u\|_{H^1} \|v\|_{H^1}, \quad \forall u, v \in H_D^1(\Omega), \quad (7)$$

where c_1 and c_2 are two generic positive constants which are independent of u, v . Let $\|\cdot\|_E = [A(\cdot, \cdot)]^{1/2}$ denote the so-called *energy norm*, then (6) and (7) guarantee that $\|\cdot\|_E$ is equivalent to $\|\cdot\|_{H^1(\Omega)}$ and therefore the variational problem (3) has a unique weak solution u in $H_D^1(\Omega)$ [14] by the Lax–Milgram Theorem.

2. The bilinear covolume-upwind finite volume approximation

For simplicity let us consider the case that Ω being a rectangular domain, $\Omega = [x_l, x_r] \times [y_b, y_t]$. Define the following partitions of Ω on each direction:

$$\begin{aligned} x_l &= x_0 < x_1 < \cdots < x_i < \cdots < x_{N_x} = x_r, \\ y_b &= y_0 < y_1 < \cdots < y_i < \cdots < y_{N_y} = y_t, \end{aligned}$$

where the partitions are not necessary to be uniform, N_x and N_y denote the numbers of grid blocks in the x and y directions, respectively. Let P_{ij} denote the grid point whose coordinate is (x_i, y_j) . Set $\Delta x_i = x_i - x_{i-1}$ for $i = 1, \dots, N_x$ and $\Delta y_j = y_j - y_{j-1}$ for $j = 1, \dots, N_y$. Then we define the primary grid Ω_h for our finite volume approximation to be the rectangular triangulation of the Ω , i.e., the set of rectangular cells

$$\{\Omega_{ij} | \Omega_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j], \quad i = 1, \dots, N_x, \quad j = 1, \dots, N_y\}.$$

In the following we will present the covolume-upwind finite volume discretization using bilinear basis functions for solving the model problem (1).

2.1. Construction of upwind control volumes

To construct standard center-based control volumes for the primary grid Ω_h , one just simply connects centers of the rectangular cells. It is well known that in the *upwind scheme* one makes use of grid points from the upwind sides during local discretizations together with the standard control volumes while the average of point values from both sides is used in the *central scheme*. Both strategies have their advantages and disadvantages as discussed in the previous section. In the proposed bilinear finite volume method we would like to apply the upwind idea only to construction of the nonstandard control volumes while the local integration for mass lumping still uses high-order quadratures such as the middle point rule used in the central scheme.

We roughly mimic the approach proposed in [16], which takes the upwind idea into consideration for construction of control volumes. The basic goal is to make the control volume for each of the grid points closer to its upwind side neighbors than to its downwind side neighbors. Instead of connecting midpoints of edges, we generate the nonstandard control volumes by connecting *optimal upwind points* on edges, which will be described below.

For each grid point we need to decide the x - and y -upwind sides by the convection term $\vec{b}(\mathbf{x}) = (b_x(\mathbf{x}), b_y(\mathbf{x}))^T$. We first take the x -direction for an illustration. Let $H(t)$ denote the Heaviside function, i.e.,

$$H(t) = \int_{-\infty}^t \delta(s) \, ds = \begin{cases} 0, & \text{if } t < 0, \\ 1, & \text{if } t \geq 0, \end{cases}$$

where $\delta(\cdot)$ denotes the Dirac function. Then, on the edge $\overline{P_{ij}P_{i+1,j}}$, we can evaluate

$$H\left(\int_{\overline{P_{ij}P_{i+1,j}}} \vec{b}(\mathbf{x}) \cdot (1, 0)^T \, ds\right) = H\left(\int_{\overline{P_{ij}P_{i+1,j}}} b_x(\mathbf{x}) \, ds\right)$$

to indicate the horizontal upwind direction. Then the x -coordinate of the *upwind point* $x_{i+1/2,j}^*$ on $\overline{P_{ij}P_{i+1,j}}$, for $i = 0, 1, \dots, N_x - 1$ and $j = 0, 1, \dots, N_y$ is defined as follows:

$$x_{i+1/2,j}^* = H\left(\int_{\overline{P_{ij}P_{i+1,j}}} b_x \, ds\right) \left[\alpha_{i+1/2,j}^x x_i + (1 - \alpha_{i+1/2,j}^x) x_{i+1} \right] + \left(1 - H\left(\int_{\overline{P_{ij}P_{i+1,j}}} b_x \, ds\right) \right) \left[(1 - \alpha_{i+1/2,j}^x) x_i + \alpha_{i+1/2,j}^x x_{i+1} \right], \quad (8)$$

where $\alpha_{i+1/2,j}^x$ is a local weight factor [29,2,19], then we obtain the upwind points $(x_{i+1/2,j}^*, y_j)$ on $\overline{P_{ij}P_{i+1,j}}$.

To decide the local weight factor, let $Pe_{i+1/2,j}^x$ denote the local Peclet's number on $\overline{P_{ij}P_{i+1,j}}$ defined by

$$Pe_{i+1/2,j}^x = \max_{x_i \leq x \leq x_{i+1}} \frac{|b_x(x, y_j)|}{a(x, y_j)} \Delta x_{i+1}. \quad (9)$$

We suggest setting

$$\alpha_{i+1/2,j}^x = \begin{cases} 1/2, & \text{if } Pe_{i+1/2,j}^x \leq 2, \\ 1 - \frac{1}{Pe_{i+1/2,j}^x}, & \text{otherwise.} \end{cases} \quad (10)$$

This idea can be derived from the one dimensional analysis of the adaptive upwind scheme [20].

Remark. There are also some other methods in the literature proposed for determining the weight factor. For example, it was proposed in [16] that

$$\alpha_{i+1/2,j}^x = \begin{cases} 1/2, & \text{if } Pe_{i+1/2,j}^x = 0, \\ \frac{1}{2} - \frac{1}{Pe_{i+1/2,j}^x} + \sqrt{\frac{1}{4} + \frac{1}{(Pe_{i+1/2,j}^x)^2}}, & \text{otherwise.} \end{cases} \quad (11)$$

Detailed discussions on reasons of this choice can be found in [16]. Other possible choices include

$$\alpha_{i+1/2,j}^x = \frac{1}{2 + Pe_{i+1/2,j}^x} \quad (12)$$

and

$$\alpha_{i+1/2,j}^x = \begin{cases} 1/2, & \text{if } Pe_{i+1/2,j}^x = 0, \\ \frac{1}{1 - e^{-Pe_{i+1/2,j}^x}} - \frac{1}{Pe_{i+1/2,j}^x}, & \text{otherwise.} \end{cases} \quad (13)$$

We note that all these methods in fact give us different but very close values for $\alpha_{i+1/2,j}^x$ when $Pe_{i+1/2,j}^x$ is large.

Similarly, we can define the upwind point $y_{ij+1/2}^*$ on $\overline{P_{ij}P_{i,j+1}}$ along y direction, for $i = 0, 1, \dots, N_x$ and $j = 0, 1, \dots, N_y - 1$ as

$$y_{ij+1/2}^* = H \left(\int_{\overline{P_{ij}P_{i,j+1}}} b_y ds \right) \left[\alpha_{ij+1/2}^y y_j + (1 - \alpha_{ij+1/2}^y) y_{j+1} \right] + \left(1 - H \left(\int_{\overline{P_{ij}P_{i,j+1}}} b_y ds \right) \right) \left[(1 - \alpha_{ij+1/2}^y) y_j + \alpha_{ij+1/2}^y y_{j+1} \right], \quad (14)$$

where

$$\alpha_{ij+1/2}^y = \begin{cases} 1/2, & \text{if } Pe_{ij+1/2}^y \leq 2, \\ 1 - \frac{1}{Pe_{ij+1/2}^y}, & \text{otherwise,} \end{cases} \quad (15)$$

with

$$Pe_{ij+1/2}^y = \max_{y_j \leq y \leq y_{j+1}} \frac{|b_y(x_i, y)|}{a(x_i, y)} \Delta y_{j+1}, \quad (16)$$

then we get the upwind point $(x_i, y_{ij+1/2}^*)$ on $\overline{P_{ij}P_{i,j+1}}$.

In each of the grid cell Ω_{ij} , we average the upwind points on opposite edges to obtain a point Q_{ij} inside the grid cell Ω_{ij} as

$$Q_{ij} = \left(\frac{x_{i-1/2,j}^* + x_{i+1/2,j-1}^*}{2}, \frac{y_{i-1,j-1/2}^* + y_{ij-1/2}^*}{2} \right), \quad (17)$$

which will be a vertex of the nonstandard control volume. Clearly, determination of the upwind points and the points Q_{ij} 's presented above can be easily extended to the three dimensional cases.

For each grid point P_{ij} , we find such Q_{ij} 's on all its four neighbor cells and connect them to form its control volume, which we denote as Ω_{ij}^* . When P_{ij} is an interior grid point, the quadrilateral $Q_{ij}Q_{i+1,j}Q_{i+1,j+1}Q_{i,j+1}$ (or $Q_1Q_2Q_3Q_4$ as shown in Fig. 1 (left)) is its upwind control volume; when P_{ij} is on the boundary, say horizontal boundary, then the quadrilateral $x_{i-1/2,j}^*x_{i+1/2,j}^*Q_{i+1,j+1}Q_{i,j+1}$ (or $Q_1Q_2Q_3Q_4$ as in Fig. 1 (right)) is its upwind control volume. The set of all such nonstandard control volumes (covolumes) $\Omega_h^* = \{\Omega_{ij}^* | i = 0, 1, \dots, N_x, j = 0, 1, \dots, N_y\}$ form a upwind dual partition of Ω with respect to the primary rectangular grid Ω_h . In contrast with the central dual decomposition, the vertices Q_{ij} 's are generally not the centers of the rectangular cells except for diffusion-dominated problems (i.e., the Peclet's numbers are all smaller than and equal to 2, in which the proposed discretization introduced below will also automatically reduce to the central scheme). It is also worth noting that if the convection coefficient is smooth, then shapes of the control volumes will be much closer to a rectangle when the grid size gets smaller.

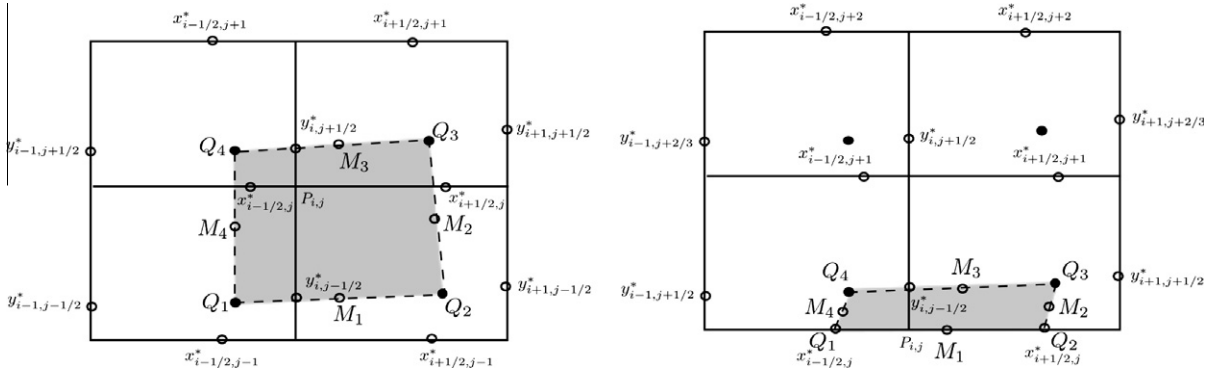


Fig. 1. The upwind control volumes associated with the grid point P_{ij} for the bilinear covolume-upwind finite volume approximation. Left: the case of P_{ij} being an interior grid point; right: the case of P_{ij} being a boundary grid point.

2.2. Bilinear finite volume discretization

With the upwind control volumes constructed above, we can now introduce bilinear finite volume discretization of the model problem (1). Choose the trial space U_h to be the piecewise bilinear function space over the rectangular grid Ω_h , i.e.,

$$U_h = \{u_h \in C(\bar{\Omega}) | u_h|_{\Omega_{ij}} \text{ is a bilinear function for any } \Omega_{ij} \in \Omega_h\}.$$

Define $\mathbf{P}_\Omega = \{P_{ij} | i = 0, 1, \dots, N_x, j = 0, 1, \dots, N_y\}$ and let $\{\phi_{ij}(\mathbf{x})\}_{P_{ij} \in \mathbf{P}_\Omega}$ be the set of basis functions of U_h associated with grid points in $\mathbf{P}_\Omega$, i.e., $\phi_{ij} \in U_h$ and

$$\phi_{ij}(\mathbf{x}) = \begin{cases} 1, & \mathbf{x} = P_{ij}, \\ 0, & \mathbf{x} \text{ is any of other grid points.} \end{cases} \quad (18)$$

For any $u_h \in U_h$, it can be written as a linear combination of the basis functions $\{\phi_{ij}\}_{P_{ij} \in \mathbf{P}_\Omega}$ such as

$$u_h(\mathbf{x}) = \sum_{P_{ij} \in \mathbf{P}_\Omega} u_{ij} \phi_{ij}(\mathbf{x}),$$

with $u_{ij} = u_h(P_{ij})$.

Define $\mathbf{P} = \{P_{ij} | P_{ij} \notin \Gamma_D, i = 0, 1, \dots, N_x, j = 0, 1, \dots, N_y\}$ and $\mathbf{P}_D = \mathbf{P}_\Omega \setminus \mathbf{P}$. We choose the test space V_h to be the piecewise constant function space over the dual partition Ω_h^* with some restrictions as follows:

$$V_h = \{v_h \in L^2(\bar{\Omega}) | v_h|_{\Omega_{ij}^*} = \text{constant for any } \Omega_{ij}^* \in \Omega_h^* \text{ and } v_h|_{\Omega_{ij}^*} = 0 \text{ if } P_{ij} \in \mathbf{P}_D\}.$$

Let $\psi_{ij}(\mathbf{x})$ be the characteristic function of the control volumes Ω_{ij}^* defined as

$$\psi_{ij}(\mathbf{x}) = \begin{cases} 1, & \mathbf{x} \in \Omega_{ij}^*, \\ 0, & \mathbf{x} \in \Omega / \Omega_{ij}^*, \end{cases} \quad (19)$$

then $\{\psi_{ij}(\mathbf{x})\}_{P_{ij} \in \mathbf{P}}$ forms a basis of V_h .

By multiplying any $v_h \in V_h$ on both sides of the governing Eq. (1), integrating over Ω and applying the Green's formula on each Ω_{ij}^* , then replacing u by $u_h \in U_h$, we obtain a discrete approximation of finite volume type for the problem (1): find $u_h \in U_h$ such that

$$\begin{cases} A_h(u_h, v_h) = L_h(v_h), & \forall v_h \in V_h, \\ u_h|_{\mathbf{P}_D} = g, \end{cases} \quad (20)$$

where $A_h(u_h, v_h) = a_h(u_h, v_h) + b_h(u_h, v_h) + c_h(u_h, v_h)$ with

$$a_h(u_h, v_h) = - \sum_{P_{ij} \in \mathbf{P}} v_h(P_{ij}) \int_{\partial \Omega_{ij}^* \cap \Gamma_N} (a \nabla u_h) \cdot \bar{\mathbf{n}}_{ij} ds, \quad (21)$$

$$b_h(u_h, v_h) = \sum_{P_{ij} \in \mathbf{P}} v_h(P_{ij}) \int_{\partial \Omega_{ij}^* \cap \Gamma_N^m} (\bar{\mathbf{b}} \cdot \bar{\mathbf{n}}_{ij}) u_h ds, \quad (22)$$

$$c_h(u_h, v_h) = \sum_{P_{ij} \in \mathbf{P}} v_h(P_{ij}) \int_{\Omega_{ij}^*} r u_h d\mathbf{x} \quad (23)$$

and

$$L_h(v_h) = \sum_{P_{ij} \in \mathbf{P}} v_h(P_{ij}) \left(\int_{\Omega_{ij}^*} f \, d\mathbf{x} - \int_{\partial\Omega_{ij}^* \cap \Gamma_N^{\text{in}}} h_{\text{in}} \, ds - \int_{\partial\Omega_{ij}^* \cap \Gamma_N^{\text{out}}} h_{\text{out}} \, ds \right), \quad (24)$$

where $\bar{\mathbf{n}}_{ij}$ denotes the unit outward normal along $\partial\Omega_{ij}^*$. It is obvious that $u_h|_{\mathbf{P}_0} = g$ directly means

$$u_{ij} = g(P_{ij}), \quad \forall P_{ij} \in \mathbf{P}_D. \quad (25)$$

Setting $v_h = \psi_{ij}$ in (20)–(24), we have that

$$- \int_{\partial\Omega_{ij}^* \cap \Gamma_N} (a \nabla u_h) \cdot \bar{\mathbf{n}}_{ij} \, ds + \int_{\partial\Omega_{ij}^* \cap \Gamma_N^{\text{in}}} (\vec{\mathbf{b}} \cdot \bar{\mathbf{n}}_{ij}) u_h \, ds + \int_{\Omega_{ij}^*} r u_h \, d\mathbf{x} = \text{RHS}, \quad \forall P_{ij} \in \mathbf{P} \quad (26)$$

where

$$\text{RHS} = \int_{\Omega_{ij}^*} f \, d\mathbf{x} - \int_{\partial\Omega_{ij}^* \cap \Gamma_N^{\text{in}}} h_{\text{in}} \, ds - \int_{\partial\Omega_{ij}^* \cap \Gamma_N^{\text{out}}} h_{\text{out}} \, ds.$$

For evaluation of the local integrations given in (26), we apply the second-order accurate midpoint rule for the mass lumping. Denote the control volume of P_{ij} as $Q_1 Q_2 Q_3 Q_4$ and let M_i be the midpoint of segments $\overline{Q_i Q_{i+1}}$ for $i = 1, 2, 3, 4$ and M be the center of the quadrilateral $Q_1 Q_2 Q_3 Q_4$ (see Fig. 1).

For the reaction term we always approximate it by applying the midpoint quadrature

$$\int_{\Omega_{ij}^*} r u_h \, d\mathbf{x} \approx u_h(M) \int_{Q_1 Q_2 Q_3 Q_4} r \, d\mathbf{x}. \quad (27)$$

Other terms have to be considered case by case according to the position of the node P_{ij} and the boundary conditions.

Case I. $P_{ij} \in \mathbf{P}$ is an interior point as shown in Fig. 1 (left). By applying the midpoint quadrature rule we get

$$\begin{aligned} - \int_{\partial\Omega_{ij}^* \cap \Gamma_N} (a \nabla u_h) \cdot \bar{\mathbf{n}}_{ij} \, ds &\approx -(\nabla u_h(M_1) \cdot \bar{\mathbf{n}}_{Q_1 Q_2}) \int_{Q_1 Q_2} a \, ds - (\nabla u_h(M_2) \cdot \bar{\mathbf{n}}_{Q_2 Q_3}) \int_{Q_2 Q_3} a \, ds \\ &\quad - (\nabla u_h(M_3) \cdot \bar{\mathbf{n}}_{Q_3 Q_4}) \int_{Q_3 Q_4} a \, ds - (\nabla u_h(M_4) \cdot \bar{\mathbf{n}}_{Q_4 Q_1}) \int_{Q_4 Q_1} a \, ds, \\ \int_{\partial\Omega_{ij}^* \cap \Gamma_N^{\text{in}}} (\vec{\mathbf{b}} \cdot \bar{\mathbf{n}}_{ij}) u_h \, ds &\approx u_h(M_1) \int_{Q_1 Q_2} \vec{\mathbf{b}} \cdot \bar{\mathbf{n}}_{Q_1 Q_2} \, ds + u_h(M_2) \int_{Q_2 Q_3} \vec{\mathbf{b}} \cdot \bar{\mathbf{n}}_{Q_2 Q_3} \, ds \\ &\quad + u_h(M_3) \int_{Q_3 Q_4} \vec{\mathbf{b}} \cdot \bar{\mathbf{n}}_{Q_3 Q_4} \, ds + u_h(M_4) \int_{Q_4 Q_1} \vec{\mathbf{b}} \cdot \bar{\mathbf{n}}_{Q_4 Q_1} \, ds, \\ \text{RHS} &= \int_{\Omega_{ij}^*} f \, d\mathbf{x}. \end{aligned} \quad (28)$$

Case II. $P_{ij} \in \mathbf{P}$ is a boundary point as shown in Fig. 1 (right) and is on Γ_N^{in} . Then we similarly get

$$\begin{aligned} - \int_{\partial\Omega_{ij}^* \cap \Gamma_N} (a \nabla u_h) \cdot \bar{\mathbf{n}}_{ij} \, ds &\approx -(\nabla u_h(M_2) \cdot \bar{\mathbf{n}}_{Q_2 Q_3}) \int_{Q_2 Q_3} a \, ds - (\nabla u_h(M_3) \cdot \bar{\mathbf{n}}_{Q_3 Q_4}) \int_{Q_3 Q_4} a \, ds - (\nabla u_h(M_4) \cdot \bar{\mathbf{n}}_{Q_4 Q_1}) \int_{Q_4 Q_1} a \, ds, \\ \int_{\partial\Omega_{ij}^* \cap \Gamma_N^{\text{in}}} (\vec{\mathbf{b}} \cdot \bar{\mathbf{n}}_{ij}) u_h \, ds &\approx u_h(M_2) \int_{Q_2 Q_3} \vec{\mathbf{b}} \cdot \bar{\mathbf{n}}_{Q_2 Q_3} \, ds + u_h(M_3) \int_{Q_3 Q_4} \vec{\mathbf{b}} \cdot \bar{\mathbf{n}}_{Q_3 Q_4} \, ds + u_h(M_4) \int_{Q_4 Q_1} \vec{\mathbf{b}} \cdot \bar{\mathbf{n}}_{Q_4 Q_1} \, ds, \\ \text{RHS} &= \int_{\Omega_{ij}^*} f \, d\mathbf{x} - \int_{Q_1 Q_2} h_{\text{in}} \, ds. \end{aligned} \quad (29)$$

Case III. $P_{ij} \in \mathbf{P}$ is a boundary point as shown in Fig. 1 (right) but belongs to Γ_N^{out} . Then we get

$$\begin{aligned} - \int_{\partial\Omega_{ij}^* \cap \Gamma_N} (a \nabla u_h) \cdot \bar{\mathbf{n}}_{ij} \, ds &\approx -(\nabla u_h(M_2) \cdot \bar{\mathbf{n}}_{Q_2 Q_3}) \int_{Q_2 Q_3} a \, ds - (\nabla u_h(M_3) \cdot \bar{\mathbf{n}}_{Q_3 Q_4}) \int_{Q_3 Q_4} a \, ds - (\nabla u_h(M_4) \cdot \bar{\mathbf{n}}_{Q_4 Q_1}) \int_{Q_4 Q_1} a \, ds, \\ \int_{\partial\Omega_{ij}^* \cap \Gamma_N^{\text{in}}} (\vec{\mathbf{b}} \cdot \bar{\mathbf{n}}_{ij}) u_h \, ds &\approx u_h(M_1) \int_{Q_1 Q_2} \vec{\mathbf{b}} \cdot \bar{\mathbf{n}}_{Q_1 Q_2} \, ds + u_h(M_2) \int_{Q_2 Q_3} \vec{\mathbf{b}} \cdot \bar{\mathbf{n}}_{Q_2 Q_3} \, ds \\ &\quad + u_h(M_3) \int_{Q_3 Q_4} \vec{\mathbf{b}} \cdot \bar{\mathbf{n}}_{Q_3 Q_4} \, ds + u_h(M_4) \int_{Q_4 Q_1} \vec{\mathbf{b}} \cdot \bar{\mathbf{n}}_{Q_4 Q_1} \, ds, \\ \text{RHS} &= \int_{\Omega_{ij}^*} f \, d\mathbf{x} - \int_{Q_1 Q_2} h_{\text{out}} \, ds. \end{aligned} \quad (30)$$

Now we need evaluate $u_h(M)$ and $\nabla u_h(M_k)$ for $k = 1, 2, 3, 4$. It is clear M_i must fall into one of the four neighbor rectangular cells around P_{ij} . Assume that $M_k = (\hat{x}, \hat{y})$ is a point inside the rectangle $P_{ij}P_{i-1,j}P_{i-1,j-1}P_{i,j-1}$ for some $(\hat{i}, \hat{j})$ with $1 \leq \hat{i} \leq N_x$, $1 \leq \hat{j} \leq N_y$. Define

$$\gamma_x = \frac{\hat{x} - x_{i-1}}{\Delta x_{i-1}}, \quad \gamma_y = \frac{\hat{y} - y_{j-1}}{\Delta y_{j-1}},$$

then we can obtain the following formulas by using the bilinear interpolation properties:

$$u_h(M_k) = (1 - \gamma_x)(1 - \gamma_y)u_{i-1,j-1} + \gamma_x\gamma_y u_{ij} + \gamma_x(1 - \gamma_y)u_{i,j-1} + (1 - \gamma_x)\gamma_y u_{i-1,j}, \quad (31)$$

$$\frac{\partial u_h}{\partial x}(M_k) = \frac{1}{\Delta x_{i-1}} (\gamma_y u_{ij} + (1 - \gamma_y)u_{i,j-1} - \gamma_y u_{i-1,j} - (1 - \gamma_y)u_{i-1,j-1}), \quad (32)$$

$$\frac{\partial u_h}{\partial y}(M_k) = \frac{1}{\Delta y_{j-1}} (\gamma_x u_{ij} + (1 - \gamma_x)u_{i-1,j} - \gamma_x u_{i,j-1} - (1 - \gamma_x)u_{i-1,j-1}). \quad (33)$$

Plugging (31)–(33) into the above equations (27)–(30) results a linear system in terms of $\{u_{ij}\}_{P_{ij} \in \mathbf{P}}$ that is the fully discrete representation of the finite volume scheme (26).

Remark. It is obvious that the proposed bilinear covolume-upwind finite volume approximation will coincide with the standard central finite volume scheme when the Peclet's numbers are all smaller than or equal to 2 in the problem (1), but it will be significantly different from the standard central or upwind schemes when the problem becomes convection-dominated (i.e., the Peclet's numbers are large).

2.3. Stability analysis

For convection-dominated problems, where the convection term dominates the diffusion term, i.e., $|\vec{\mathbf{b}}| \gg a$, it is well known that large nonphysical oscillations will appear in the numerical solution when the problem is solved by the standard central scheme. The upwind finite volume scheme could overcome this numerical instability issue, while its convergence would be compromised due to the use of information from only the upwind sides. The proposed bilinear covolume-upwind finite volume scheme is actually a modified central scheme (still use the midpoint quadrature) where the Peclet-number information is used for generating the upwind control volumes. This scheme basically has second-order local truncation errors as the central scheme when the grid size is small, thus is expected to be of second order accuracy in the L^2 norm and of first order in the H^1 norm if it is stable. In this section, we will analyze its stability for solving the model problem with constant coefficients.

Without loss of generality, we assume Ω_h consists of uniform rectangles, i.e. $\Delta x_i = h_x$ and $\Delta y_j = h_y$ for all i and j . We also restrain our discussion within the problem (1) with a pure Dirichlet boundary condition in which the diffusion coefficient a and the convection wind $\vec{\mathbf{b}} = (b_x, b_y)$ are constants, and a reaction term $r = 0$. Thus we know $Pe_{i+1/2,j}^x \equiv Pe^x$, $\alpha_{i+1/2,j}^x \equiv \alpha^x$, $Pe_{i,j+1/2}^y \equiv Pe^y$ and $\alpha_{i,j+1/2}^y \equiv \alpha^y$ for some constants Pe^x , Pe^y , α^x and α^y .

We may further assume that $b_x \geq 0$ and $b_y \geq 0$, then it is easy to find that the control volume Ω_{ij}^* 's are uniform rectangles and left-bottom (upstream) shifted from the standard centered ones, see Fig. 2 for description. Under this assumption it is clear that

$$Pe^x = \frac{b_x h_x}{a}, \quad Pe^y = \frac{b_y h_y}{a} \quad (34)$$

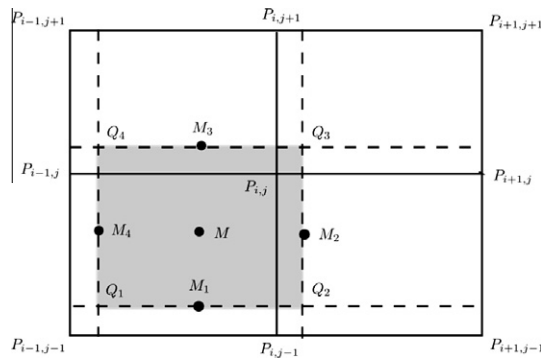


Fig. 2. The upwind control volumes for the problem with constant diffusion–convection coefficients in the bilinear covolume-upwind finite volume scheme.

and we can compute γ_x and γ_y for M_i 's as follows:

$$\begin{aligned}\gamma_x(M_1) &= \gamma_x(M_3) = \frac{3}{2} - \alpha^x, & \gamma_y(M_1) &= \gamma_y(M_3) = 1 - \alpha^y, \\ \gamma_x(M_2) &= \gamma_x(M_4) = 1 - \alpha^x, & \gamma_y(M_2) &= \gamma_y(M_4) = \frac{3}{2} - \alpha^y.\end{aligned}$$

Put $\{\gamma_x(M_i), \gamma_y(M_i)\}$ into (31)–(33), then the discretized system (26) can be written as the following equations: for any $P_{ij} \in \mathbf{P}$,

$$C_1 u_{i-1,j-1} + C_2 u_{i-1,j} + C_3 u_{i-1,j+1} + C_4 u_{i,j-1} + C_5 u_{i,j} + C_6 u_{i,j+1} + C_7 u_{i+1,j-1} + C_8 u_{i+1,j} + C_9 u_{i+1,j+1} = F_{ij}, \quad (35)$$

where $\{C_i\}_{i=1}^9$ are some coefficients determined by a , b_x , b_y , h_x , h_y , with $\sum_{k=1}^8 C_k = 0$ and $C_9 = 0$.

Since the proposed scheme is identical to the standard central scheme when $Pe^x \leq 2$ and $Pe^y \leq 2$ that is well known to be stable in this case, thus we confine our analysis to the situations of $Pe^x > 2$ and/or $Pe^y > 2$.

Case I. Only one of the Pe^x and Pe^y is greater than 2. This generally means the wind is strong in one direction and very mild in the other. Without loss of generality, let us assume $Pe^x > 2$ and $Pe^y \leq 2$. By (10) and (15) we have

$$\alpha^x = 1 - \frac{a}{b_x h_x}, \quad \alpha^y = 1/2. \quad (36)$$

We then can obtain that

$$\begin{aligned}C_1 &= -\frac{(b_x h_x - 2a)(2a + b_y h_y)}{4b_x h_y}, & C_2 &= -\frac{2a^2}{b_x h_y} - b_x h_y + \frac{a h_x}{h_y}, \\ C_3 &= -\frac{(b_x h_x - 2a)(2a - b_y h_y)}{4b_x h_y}, & C_4 &= -\frac{(b_x h_x + 2a)(2a + b_y h_y)}{4b_x h_y}, \\ C_5 &= \frac{a h_x}{h_y} + \frac{2a^2}{b_x h_y} + b_x h_y, & C_6 &= -\frac{(b_x h_x + 2a)(2a - b_y h_y)}{4b_x h_y}, \\ C_7 &= C_8 = 0.\end{aligned}$$

It is easy to verify that

$$\begin{aligned}C_1 &= -\frac{a^2(Pe_x - 2)(2 + Pe_y)}{4b_x h_y} < 0, & C_2 &= -\frac{b_x}{h_y} \left(h_y^2 - h_x^2 \frac{Pe_x - 2}{Pe_x^2} \right), \\ C_3 &= -\frac{a^2(Pe_x - 2)(2 - Pe_y)}{4b_x h_y} \leq 0, & C_4 &= -\frac{a^2(Pe_x + 2)(2 + Pe_y)}{4b_x h_y} < 0, \\ C_5 &> 0, & C_6 &= -\frac{a^2(Pe_x + 2)(2 - Pe_y)}{4b_x h_y} \leq 0.\end{aligned}$$

If we impose a restriction on selection of the mesh size such as

$$h_y \geq h_x \frac{\sqrt{Pe_x - 2}}{Pe_x}, \quad (37)$$

then it also holds $C_2 \leq 0$ and consequently the system (35) results in an M -matrix (a common characterization of an M -matrix is a diagonally dominated square matrix with non-positive off-diagonal entries and positive diagonal entries). Thus the proposed scheme satisfies the *maximum principle* (consequently stable) with such a restriction. It is not difficult to find that $\max_{Pe_x \geq 2} \frac{\sqrt{Pe_x - 2}}{Pe_x} = \sqrt{2}/4$ and $\lim_{Pe_x \rightarrow \infty} \frac{\sqrt{Pe_x - 2}}{Pe_x} = 0$, thus the condition (37) can be easily satisfied by choosing approximate h_x and h_y .

Case II. Both of Pe_x and Pe_y are greater than 2. Then it holds

$$\alpha^x = 1 - \frac{a}{b_x h_x}, \quad \alpha^y = 1 - \frac{a}{b_y h_y} \quad (38)$$

and we have that

$$\begin{aligned}C_1 &= -\frac{1}{2}(b_x h_y + b_y h_x) + a \left(\frac{b_y}{b_x} + \frac{b_x}{b_y} \right), & C_2 &= -\frac{1}{2}(b_x h_y - b_y h_x) - a \left(\frac{b_y}{b_x} + \frac{b_x}{b_y} \right), \\ C_3 &= 0, & C_4 &= -\frac{1}{2}(b_y h_x - b_x h_y) - a \left(\frac{b_y}{b_x} + \frac{b_x}{b_y} \right), \\ C_5 &= \frac{1}{2}(b_y h_x + b_x h_y) + a \left(\frac{b_y}{b_x} + \frac{b_x}{b_y} \right), & C_6 &= C_7 = C_8 = 0.\end{aligned}$$

If the unknowns u_{ij} 's are ordered first in x -direction then in y -direction, then the system (35) obviously gives us a lower triangular system with positive diagonal elements since $C_5 > 0$. It is also clear that the diagonal component $C_5 > |C_k|$ for

$k = 1, 2, 4$ (since $a > 0$ and $b_x, b_y > 0$ in this case), so the system is stable since all eigenvalues of the system are strictly positive, but it still does not satisfy the maximum principle in general.

We further can find that

$$C_1 = -\frac{ab_y(Pe_x - 2)}{2b_x} - \frac{ab_x(Pe_y - 2)}{2b_y} < 0.$$

If we additionally impose a restriction on selection of the mesh size such as

$$\left| \frac{h_x}{b_x} - \frac{h_y}{b_y} \right| \leq 2a \left(\frac{1}{b_x^2} + \frac{1}{b_y^2} \right), \quad (39)$$

then we can get

$$C_2 \leq 0, \quad C_4 \leq 0.$$

Thus in this situation the system (35) also results in an M -matrix, and consequently the proposed scheme satisfies the *maximum principle*. A simple way to enforce the relation (39) in practical numerical simulations is to set the directional ratio of the mesh sizes h_x/h_y equal to the directional ratio of the wind amplitudes b_x/b_y (this strategy is used later in our numerical experiments presented in Section 4).

Remark. We have so far discussed numerical stability of the proposed bilinear covolume-upwind finite volume approximation for the model problem with constant coefficients. When the discretization is stable, in combination with the fact of consistency that the local truncation errors are at least of the second order in the bilinear CUFVM, it is then easy to obtain convergence of the method to the exact solution by the Lax Equivalence Theorem [15].

The cases of general variable coefficients are still under our investigation; on the other hand, we will numerically demonstrate in Section 4 the stability and optimal convergence of the proposed methods by various examples.

3. The biquadratic covolume-upwind finite volume approximation

In this section, we will extend the idea of construction of upwind control volumes and develop a covolume-upwind finite volume scheme using biquadratic basis functions for the model problem (1), which is expected to have one-order higher convergence than the previous one.

For bilinear basis functions, we have four degrees of freedom (DOF), the four grid points in each rectangular cell; while for biquadratic basis functions, we will have nine DOF in each grid cell, which include the four grid points, four midpoints of its edges and the center of the grid cell. According to the classical finite volume theory, we need to construct control volumes for each of the DOF [17].

3.1. Construction of upwind control volumes

For each cell in Ω_h , by connecting *midpoints* of opposite sides, we divide the primary grid cell $\Omega_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j]$ into four equal-sized rectangular subcells. As shown in Fig. 3, in the primary cell $\Omega_{i+1/2,j+1/2}$, define $P_{i+1/2,j} = (x_{i+1/2}, y_j)$, $P_{i,j+1/2} = (x_i, y_{j+1/2})$, $P_{i+1,j+1/2} = (x_{i+1}, y_{j+1/2})$ and $P_{i+1/2,j+1} = (x_{i+1/2}, y_{j+1})$ to be the midpoints of its edges and $P_{i+1/2,j+1/2} = (x_{i+1/2}, y_{j+1/2})$ to be its center where $x_{i+1/2} = (x_i + x_{i+1})/2$, $y_{j+1/2} = (y_j + y_{j+1})/2$ for $i = 0, 1, \dots, N_x - 1$, $j = 0, 1, \dots, N_y - 1$.

Let us first define two upwind points on each of the four edges of $\Omega_{i+1/2,j+1/2}$. The way we determine the two upwind points is similar to that in Section 2 but using the subcells instead of the primary grid cells. We define x -coordinates of the upwind points $(x_{i+1/4,j}^*, y_j)$ and $(x_{i+3/4,j}^*, y_j)$ on $\overline{P_{ij}P_{i+1,j}}$, for $i = 0, 1, \dots, N_x - 1$ and $j = 0, 1, \dots, N_y$ as follows

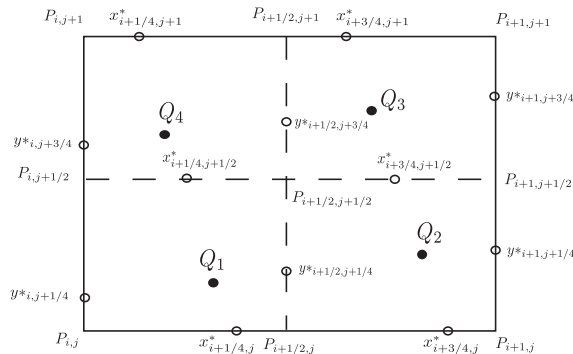


Fig. 3. The subcells and the upwind points of a grid cell for biquadratic basis functions.

$$x_{i+1/4,j}^* = H\left(\int_{\overline{P_{ij}P_{i+1,j}}} b_x ds\right) \left[\alpha_{i+1/2,j}^x x_i + (1 - \alpha_{i+1/2,j}^x) x_{i+1} \right] + \left(1 - H\left(\int_{\overline{P_{ij}P_{i+1,j}}} b_x ds\right) \right) \left[(1 - \alpha_{i+1/2,j}^x) x_i + \alpha_{i+1/2,j}^x x_{i+1/2} \right], \quad (40)$$

$$x_{i+3/4,j}^* = H\left(\int_{\overline{P_{ij}P_{i+1,j}}} b_x ds\right) \left[\alpha_{i+1/2,j}^x x_{i+1/2} + (1 - \alpha_{i+1/2,j}^x) x_{i+1} \right] + \left(1 - H\left(\int_{\overline{P_{ij}P_{i+1,j}}} b_x ds\right) \right) \left[(1 - \alpha_{i+1/2,j}^x) x_{i+1/2} + \alpha_{i+1/2,j}^x x_{i+1} \right], \quad (41)$$

where $\alpha_{i+1/2,j}^x$ is the local weight factor associated with $\overline{P_{ij}P_{i+1,j}}$ calculated by (10). Similarly we can define the upwind points $(x_i, y_{i,j+1/4}^*)$ and $(x_i, y_{i,j+3/4}^*)$ on $\overline{P_{ij}P_{i+1,j}}$, for $i = 0, 1, \dots, N_x$ and $j = 0, 1, \dots, N_y - 1$ as follows

$$y_{i,j+1/4}^* = H\left(\int_{\overline{P_{ij}P_{i+1,j}}} b_y ds\right) \left[\alpha_{i,j+1/2}^y y_j + (1 - \alpha_{i,j+1/2}^y) y_{j+1} \right] + \left(1 - H\left(\int_{\overline{P_{ij}P_{i+1,j}}} b_y ds\right) \right) \left[(1 - \alpha_{i,j+1/2}^y) y_j + \alpha_{i,j+1/2}^y y_{j+1/2} \right], \quad (42)$$

$$y_{i,j+3/4}^* = H\left(\int_{\overline{P_{ij}P_{i+1,j}}} b_y ds\right) \left[\alpha_{i,j+1/2}^y y_{j+1/2} + (1 - \alpha_{i,j+1/2}^y) y_{j+1} \right] + \left(1 - H\left(\int_{\overline{P_{ij}P_{i+1,j}}} b_y ds\right) \right) \left[(1 - \alpha_{i,j+1/2}^y) y_{j+1/2} + \alpha_{i,j+1/2}^y y_{j+1} \right], \quad (43)$$

where $\alpha_{i,j+1/2}^y$ is the local weight factor associated with $\overline{P_{ij}P_{i+1,j}}$ defined in (15).

Now we continue to define the remaining four upwind points inside the grid cell $\Omega_{i+1,j+1}$ so that all side of its four subcells has a upwind points as shown in Fig. 3. We set the point $(x_{i+1/4,j+1/2}^*, y_{j+1/2}^*)$ to be the intersection of $(x_{i+1/4,j}^*, y_j^*)$ and $\overline{P_{i+1/2,j}P_{i+1/2,j+1}}$, thus

$$x_{i+1/4,j+1/2}^* = (x_{i+1/4,j}^* + x_{i+1/4,j+1}^*)/2$$

and similarly

$$x_{i+3/4,j+1/2}^* = (x_{i+3/4,j}^* + x_{i+3/4,j+1}^*)/2.$$

Then we define $(x_{i+1/2,j+1/4}^*, y_{j+1/4}^*)$ to be the intersection of $(x_i, y_{i,j+1/4}^*)$ and $\overline{P_{i+1/2,j}P_{i+1/2,j+1}}$, thus

$$y_{i+1/2,j+1/4}^* = (y_{i,j+1/4}^* + y_{i+1,j+1/4}^*)/2$$

and similarly

$$y_{i+1/2,j+3/4}^* = (y_{i,j+3/4}^* + y_{i+1,j+3/4}^*)/2.$$

For every primary cell $\Omega_{i+1,j+1}$, we determine its four points Q_1 , Q_2 , Q_3 and Q_4 by setting

$$\begin{aligned} Q_{i+1,j+1}^1 &= \left(\frac{x_{i+1/4,j}^* + x_{i+1/4,j+1/2}^*}{2}, \frac{y_{i,j+1/4}^* + y_{i+1/2,j+1/4}^*}{2} \right), \\ Q_{i+1,j+1}^2 &= \left(\frac{x_{i+3/4,j}^* + x_{i+3/4,j+1/2}^*}{2}, \frac{y_{i+1/2,j+1/4}^* + y_{i+1,j+1/4}^*}{2} \right), \\ Q_{i+1,j+1}^3 &= \left(\frac{x_{i+3/4,j+1/2}^* + x_{i+3/4,j+1}^*}{2}, \frac{y_{i+1/2,j+3/4}^* + y_{i+1,j+3/4}^*}{2} \right), \\ Q_{i+1,j+1}^4 &= \left(\frac{x_{i+1/4,j+1/2}^* + x_{i+1/4,j+1}^*}{2}, \frac{y_{i,j+3/4}^* + y_{i+1/2,j+3/4}^*}{2} \right). \end{aligned} \quad (44)$$

Clearly each subcell of $\Omega_{i+1,j+1}$ contains exactly one of the four points as denoted by filled circles in Fig. 4 and Fig. 5. We will use these points to determine the upwind control volumes for DOF of the biquadratic basis functions.

For each DOF of biquadratic basis functions, we define its control volume as the quadrilateral formed by connecting the Q_i points in its four neighbor subcells counterclockwisely. For better understanding, the typical upwind control volumes for interior DOFs are shown in Fig. 4, while those for boundary DOFs are presented in Fig. 5. For primary grid point P_{ij} , let us denote its control volume as $\tilde{\Omega}_{ij}^*$; for nodes $P_{i+1/2,j}$, $P_{i,j+1/2}$ and $P_{i+1/2,j+1/2}$, denote their control volumes as $\tilde{\Omega}_{i+1/2,j}^*$, $\tilde{\Omega}_{i,j+1/2}^*$ and $\tilde{\Omega}_{i+1/2,j+1/2}^*$ respectively for proper i 's and j 's. Then the nonstandard up-wind dual partition of the domain Ω associated with biquadratic basis functions is given by

$$\tilde{\Omega}_h^* = \{ \tilde{\Omega}_{ij}^*, \tilde{\Omega}_{i+1/2,j}^*, \tilde{\Omega}_{i,j+1/2}^*, \tilde{\Omega}_{i+1/2,j+1/2}^* \mid i = 0, 1, \dots, N_x - 1, j = 0, 1, \dots, N_y \} \cup \{ \Omega_{N_x, N_y}^* \}. \quad (45)$$

3.2. Biquadratic finite volume discretization

With the upwind control volumes just constructed, we now can introduce the biquadratic finite volume discretization of the model problem (1). We choose the trial space $\tilde{U}_h$ to be the piecewise biquadratic function space over the rectangular decomposition of Ω_h , i.e.

$$\tilde{U}_h = \{ u_h \in C(\bar{\Omega}) \mid u_h|_{\Omega_{ij}} \text{ is a biquadratic function for any } \Omega_{ij} \in \Omega_h \}.$$

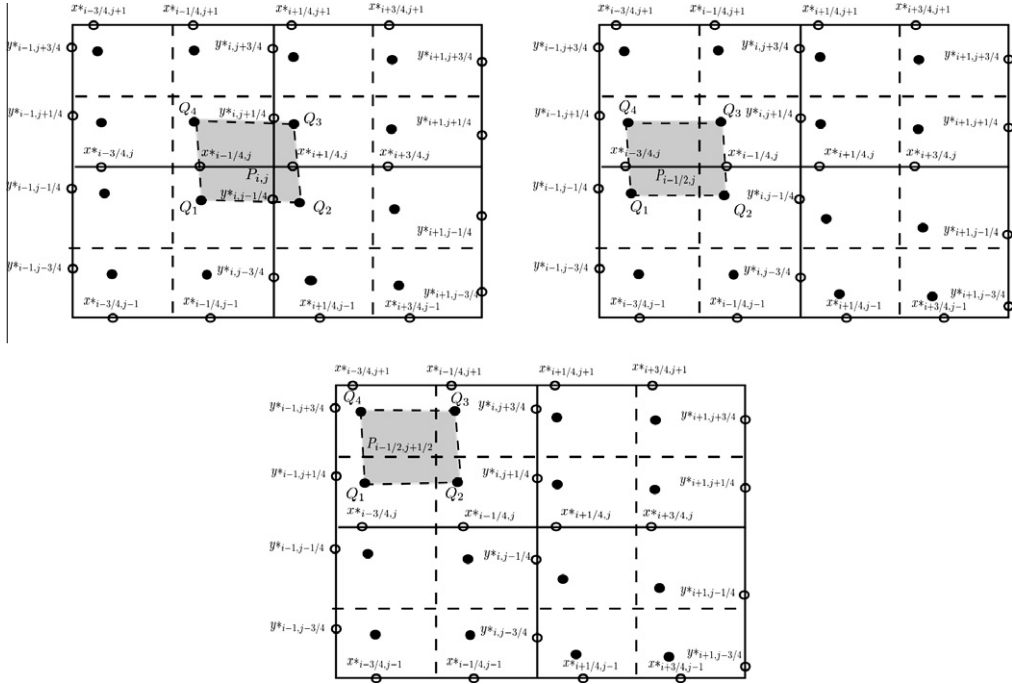


Fig. 4. The upwind control volumes associated with interior DOF for the biquadratic covolume-upwind finite volume approximation. Top-left: the DOF on a primary grid point; top-right: the DOF on the midpoint of an edge of a primary cell; bottom: the DOF on the center of a primary cell.

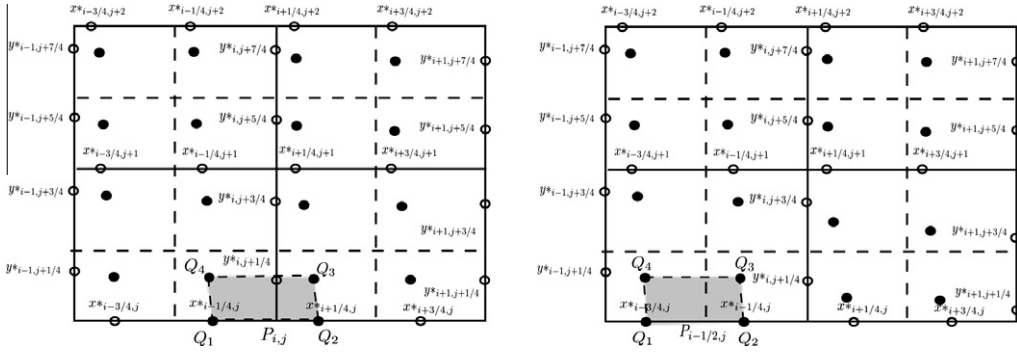


Fig. 5. The upwind control volumes associated with boundary DOF for the biquadratic covolume-upwind finite volume approximation. Left: the DOF on a primary grid point; right: the DOF on the midpoint of an edge of a primary cell.

Define $\tilde{\mathbf{P}}_\Omega = \{P_{s,l} | s = 0, \frac{1}{2}, 1, \dots, N_x - \frac{1}{2}, N_x, l = 0, \frac{1}{2}, 1, \dots, N_y - \frac{1}{2}, N_y\}$ and let $\{\phi_{s,l}(\mathbf{x})\}_{P_{s,l} \in \tilde{\mathbf{P}}_\Omega}$ be the set of basis functions of $\tilde{U}_h$ associated with points in $\tilde{\mathbf{P}}_\Omega$, i.e., $\tilde{\phi}_{s,l} \in \tilde{U}_h$ and

$$\tilde{\phi}_{s,l}(\mathbf{x}) = \begin{cases} 1, & \mathbf{x} = P_{s,l}, \\ 0, & \mathbf{x} \text{ is any of other points in } \tilde{\mathbf{P}}_\Omega. \end{cases} \quad (46)$$

Then for any $u_h \in \tilde{U}_h$, it can be written as a linear combination of the basis functions $\{\tilde{\phi}_{s,l}(\mathbf{x})\}_{P_{s,l} \in \tilde{\mathbf{P}}_\Omega}$ such as

$$u_h = \sum_{P_{s,l} \in \tilde{\mathbf{P}}} u_{s,l} \tilde{\phi}_{s,l}(\mathbf{x}),$$

where $u_{s,l} = u_h(P_{s,l})$.

Define $\tilde{\mathbf{P}} = \{P_{s,l} | P_{s,l} \notin \Gamma_D, s = 0, \frac{1}{2}, \dots, N_x - \frac{1}{2}, N_x, l = 0, \frac{1}{2}, \dots, N_y - \frac{1}{2}, N_y\}$ and $\tilde{\mathbf{P}}_D = \tilde{\mathbf{P}}_\Omega \setminus \tilde{\mathbf{P}}$. We choose the test space $\tilde{V}_h$ to be the piecewise constant function space over the dual partition $\tilde{\Omega}_h^*$ with some restrictions as follows:

$$\tilde{V}_h = \{v_h \in L^2(\tilde{\Omega}) | v_h|_{\tilde{\Omega}_{s,l}^*} = \text{constant for any } \tilde{\Omega}_{s,l}^* \text{ and } v_h|_{\tilde{\Omega}_{s,l}^*} = 0 \text{ if } P_{s,l} \in \tilde{\mathbf{P}}_D\}.$$

Let $\tilde{\psi}_{s,l}(\mathbf{x})$ be the characteristic function of control volumes $\tilde{\Omega}_{s,l}^* \in \tilde{\Omega}_h^*$, defined as

$$\tilde{\psi}_{s,l}(\mathbf{x}) = \begin{cases} 1, & \mathbf{x} \in \tilde{\Omega}_{s,l}^*, \\ 0, & \mathbf{x} \notin \tilde{\Omega}_{s,l}^*, \end{cases} \quad (47)$$

then $\{\tilde{\psi}_{s,l}\}_{P_{s,l} \in \tilde{\mathbf{P}}}$ form a basis of $\tilde{V}_h$.

The biquadratic finite volume discretization for the problem (1) is then given by: find $u_h \in \tilde{U}_h$ such that

$$\begin{cases} A_h(u_h, v_h) = L_h(v_h), & \forall v_h \in \tilde{V}_h, \\ u_h|_{\tilde{\mathbf{P}}_D} = g, \end{cases} \quad (48)$$

where A_h and L_h are the same as those defined in (21)–(24). Again $u_h|_{\tilde{\mathbf{P}}_D} = g$ directly gives us

$$u_{s,l} = g(P_{s,l}), \quad \forall P_{s,l} \in \tilde{\mathbf{P}}_D. \quad (49)$$

By setting $v_h = \tilde{\psi}_{i,j}$ in (48), we still have that for every $P_{s,l} \in \tilde{\mathbf{P}}$, we have

$$-\int_{\partial\tilde{\Omega}_{s,l}^* \setminus \Gamma_N} (a \nabla u_h) \cdot \tilde{\mathbf{n}}_{s,l} ds + \int_{\partial\tilde{\Omega}_{s,l}^* \setminus \Gamma_N^{\text{in}}} (\tilde{\mathbf{b}} \cdot \tilde{\mathbf{n}}_{s,l}) u_h ds + \int_{\tilde{\Omega}_{s,l}^*} r u_h d\mathbf{x} = \text{RHS}, \quad (50)$$

with

$$\text{RHS} = \int_{\tilde{\Omega}_{s,l}^*} f d\mathbf{x} - \int_{\partial\tilde{\Omega}_{s,l}^* \cap \Gamma_N^{\text{in}}} h_{\text{in}} ds - \int_{\partial\tilde{\Omega}_{s,l}^* \cap \Gamma_N^{\text{out}}} h_{\text{out}} ds.$$

We again apply the similar quadrature and mass lumping for evaluation of the local integrations contained in (21)–(23) and obtain (27)–(30) for approximation of (50). Now it again only remains efficient evaluation of u_h and ∇u_h at a given point. Suppose $M = (\hat{x}, \hat{y})$ is a point in the primary cell $\Omega_{i,j} = [x_{i-1}, x_i] \times [y_{j-1}, y_j]$ for some (i, j) with $1 \leq i \leq N_x$, $1 \leq j \leq N_y$. Let

$$\xi = \frac{2(\hat{x} - x_{i-\frac{1}{2}})}{\Delta x_i}, \quad \eta = \frac{2(\hat{y} - y_{j-\frac{1}{2}})}{\Delta y_j}$$

and

$$\{\xi_k\}_{k=1}^9 = \{-1, 1, 1, -1, 0, 1, 0, -1, 0\}, \quad \{\eta_k\}_{k=1}^9 = \{-1, -1, 1, 1, -1, 0, 1, 0, 0\}.$$

We can represent the nine shape functions of the biquadratic interpolation in the cell $\Omega_{i,j}$ [27] as follows:

$$\begin{aligned} \Psi_k(\xi, \eta) &= \frac{1}{4} \xi_k \eta_k \xi \eta (1 + \xi_k \xi) (1 + \eta_k \eta), \quad k = 1, 2, 3, 4, \\ \Psi_k(\xi, \eta) &= \frac{1}{2} \eta_k \eta (1 - \xi^2) (1 + \eta_k \eta), \quad k = 5, 7, \\ \Psi_k(\xi, \eta) &= \frac{1}{2} \xi_k \xi (1 + \xi_k \xi) (1 - \eta^2), \quad k = 6, 8, \\ \Psi_9(\xi, \eta) &= (1 - \xi^2) (1 - \eta^2). \end{aligned}$$

Then it is easy to verify that

$$u_h(M) = \sum_{k=1}^9 u_{i-\frac{1}{2}+\xi_k j-\frac{1}{2}+\eta_k} \Psi_k(M). \quad (51)$$

Let

$$\begin{aligned} d_x &= \hat{x} - x_{i-1} - \frac{1}{4} \Delta x_i, \quad d_y = \hat{y} - y_{j-1} - \frac{1}{4} \Delta y_j, \\ r_x &= (\hat{x} - x_{i-1}) / \Delta x_i, \quad r_y = (\hat{y} - y_{j-1}) / \Delta y_j \end{aligned}$$

and set

$$\begin{aligned} A_x &= 2(u_{i-1,j} + u_{i-1,j-1} - 2u_{i-1,j-\frac{1}{2}})r_y^2 + (4u_{i-1,j-\frac{1}{2}} - u_{i-1,j} - 3u_{i-1,j-1})r_y + u_{i-1,j-1}, \\ B_x &= 2(u_{i,j} + u_{i,j-1} - 2u_{i,j-\frac{1}{2}})r_y^2 + (4u_{i,j-\frac{1}{2}} - u_{i,j} - 3u_{i,j-1})r_y + u_{i,j-1}, \\ C_x &= 2(u_{i-\frac{1}{2},j} + u_{i-\frac{1}{2},j-1} - 2u_{i-\frac{1}{2},j-\frac{1}{2}})r_y^2 + (4u_{i-\frac{1}{2},j-\frac{1}{2}} - u_{i-\frac{1}{2},j} - 3u_{i-\frac{1}{2},j-1})r_y + u_{i-\frac{1}{2},j-1}, \\ A_y &= 2(u_{i,j-1} + u_{i-1,j-1} - 2u_{i-\frac{1}{2},j-1})r_x^2 + (4u_{i-\frac{1}{2},j-1} - u_{i,j-1} - 3u_{i-1,j-1})r_x + u_{i-1,j-1}, \\ B_y &= 2(u_{i,j} + u_{i-1,j} - 2u_{i-\frac{1}{2},j})r_x^2 + (4u_{i-\frac{1}{2},j} - u_{i,j} - 3u_{i-1,j})r_x + u_{i-1,j}, \\ C_y &= 2(u_{i,j-\frac{1}{2}} + u_{i-1,j-\frac{1}{2}} - 2u_{i-\frac{1}{2},j-\frac{1}{2}})r_x^2 + (4u_{i-\frac{1}{2},j-\frac{1}{2}} - u_{i,j-\frac{1}{2}} - 3u_{i-1,j-\frac{1}{2}})r_x + u_{i-1,j-\frac{1}{2}}. \end{aligned}$$

Then we have

$$\frac{\partial u_h}{\partial x}(M) = \frac{1}{\Delta x_i} [(-2 + 4d_x)A_x + 4d_x B_x + (2 - 8d_x)C_x], \quad (52)$$

$$\frac{\partial u_h}{\partial y}(M) = \frac{1}{\Delta y_j} [(-2 + 4d_y)A_y + 4d_y B_y + (2 - 8d_y)C_y]. \quad (53)$$

By using (51)–(53), we can express (48) as a linear system and then solve it for $\{u_{s,l}\}_{p_{s,l} \in \bar{p}}$.

Similar stability analysis as that in Section 2.3 can be done for the biquadratic covolume-upwind finite volume approximation although we skip it here since it is tedious and has no further innovation. We will demonstrate its stability and optimal convergence through extensive numerical examples in the next section.

4. Numerical experiments

In this section we will carry out various numerical experiments to test performance of the proposed bilinear and biquadratic covolume-upwind finite volume methods (CUFVM). We will demonstrate good conditioning of the discretized systems by the proposed methods through some comparisons with the standard upwind finite volume method (UFVM) and the optimal weighted upwinding covolume method (OWUCM), and disappearance of nonphysical oscillations especially for convection-dominated problems. We will also show optimal convergence of these two schemes, i.e., second-order in the L^2 norm and first-order in the H^1 norm for the bilinear CUFVM, and third-order in the L^2 norm and second-order in the H^1 norm for the biquadratic CUFVM. In addition, we will also present the superconvergence property in gradient recovery possessed by the two CUFVM methods at centers of the primary cells.

We define the following measures associated with the mesh Ω_h for the error $e_h = u - u_h$:

$$\begin{aligned} \|e_h\|_\infty &= \max |e_h|, \\ \|e_h\|_{L^2} &= \left(\sum_{ij} \int_{\Omega_{ij}} |e_h|^2 d\mathbf{x} \right)^{1/2}, \\ |e_h|_{H^1} &= \left(\sum_{ij} \int_{\Omega_{ij}} |\nabla e_h|^2 d\mathbf{x} \right)^{1/2}, \\ |e_h|_{1,\Omega_h} &= \left(\sum_{ij} |\nabla e_h(x_{i-1/2}, y_{j-1/2})|^2 \Delta x_i \Delta y_j \right)^{1/2}. \end{aligned}$$

For each example in the numerical experiments, we let the domain $\Omega = [0, 1] \times [0, 1]$, and solve the model problem (1) using the bilinear CUFVM, the biquadratic CUFVM, the UFVM respectively. Errors are computed by the above measurements together with corresponding convergence rates. We also provide condition numbers of the discretized systems to demonstrate stability and solvability of the discretized systems by the bilinear and biquadratic CUFVMs.

Example 1. In this example, we set the convection coefficient $\vec{\mathbf{b}}(\mathbf{x}) = (2, 1)$ and the reaction term $r(\mathbf{x}) = 1$ while the diffusion coefficient is chosen to be $a(\mathbf{x}) = 1, 10^{-3}, 10^{-8}$, respectively. Clearly $\nabla \cdot \mathbf{b} = 0$, thus the assumption (2) is satisfied. The exact solution is set to be

$$u(\mathbf{x}) = 1 + e^{x+y}, \quad (54)$$

which is very smooth (see Fig. 6). We test the model problem (1) with pure Dirichlet boundary condition, i.e., $\Gamma_D = \partial\Omega$ and $\Gamma_N^{\text{in}} = \Gamma_N^{\text{out}} = \emptyset$, with $g = u|_{\partial\Omega}$. The source function $f(\mathbf{x})$ is then determined by u and the left-hand side of the Eq. (1).

We first tested the bilinear CUFVM on uniform grids with sizes $N_x \times N_y = 10 \times 10, 20 \times 20, 40 \times 40$, and 80×80 . The computational results are reported in Table 1. We can clearly observe second-order convergence in $\|e_h\|_\infty$ and $\|e_h\|_{L^2}$ and first-order convergence in $|e_h|_{H^1}$ for all choices of a . The convergence rates of $|e_h|_{1,\Omega_h}$ are 2 for the case of $a = 1$ which reflect superconvergence at the centers of the grid cells, but they degenerate to values slightly above 1.5 for the convection-dominated ($a = 10^{-3}$) case and the strongly convection-dominated ($a = 10^{-8}$) case, which are still very good. Results by the biquadratic CUFVM on the same uniform grids are reported in Table 2. We obtain third-order convergence for $\|e_h\|_\infty$, $\|e_h\|_{L^2}$ and $|e_h|_{1,\Omega_h}$ and second-order convergence for $|e_h|_{H^1}$ for the cases of $a = 10^{-3}$ and $a = 10^{-8}$. The convergence rates of $\|e_h\|_\infty$ and $\|e_h\|_{L^2}$ decrease to around 2.3 from the optimal one along the grid refinements when $a = 1$, and in particular both $|e_h|_{H^1}$ and $|e_h|_{1,\Omega_h}$ are all second-order convergent in this case but the latter one has great improvements in the values (more than one magnitude). Condition numbers of the discrete systems produced by both CUFVMs increase proportionally by a rate of roughly 4 for $a = 1$ and between 4 and 7 for $a = 10^{-8}$ along the uniform grid refinements. We also observe in both methods the increasing of condition numbers slows down for $a = 10^{-3}$ along the grid refinements, which is partly due to the fact that the discrete systems gradually converge to M -matrix. In addition, the condition numbers increase very slowly along decreasing of the diffusion coefficient a . See Fig. 7 (left) for a visual illustration of these behaviors.

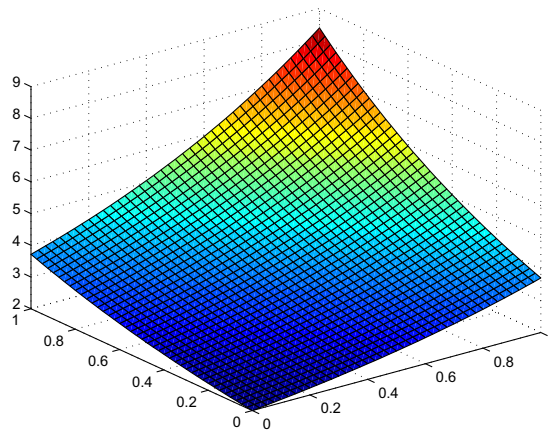


Fig. 6. Plot of the exact solution u defined in (54) in Example 1.

Table 1

Numerical results of Example 1 using the bilinear CUFVM on uniform grids.

a	$N_x \times N_y$	$\ e_h\ _\infty$	CR	$\ e_h\ _{L^2}$	CR	$ e_h _{H^1}$	CR	$ e_h _{1,\Omega_h}$	CR	Cond
1	10×10	$1.514\text{e-}2$	–	$5.684\text{e-}3$	–	$1.304\text{e-}1$	–	$7.396\text{e-}3$	–	5.589e+1
	20×20	$3.931\text{e-}3$	1.95	$1.423\text{e-}3$	2.00	$6.521\text{e-}2$	1.00	$1.849\text{e-}3$	2.00	2.276e+2
	40×40	$1.001\text{e-}4$	1.97	$3.558\text{e-}4$	2.00	$3.260\text{e-}2$	1.00	$4.623\text{e-}4$	2.00	9.173e+2
	80×80	$2.528\text{e-}4$	1.99	$8.897\text{e-}5$	2.00	$1.630\text{e-}2$	1.00	$1.155\text{e-}4$	2.00	3.680e+3
10^{-3}	10×10	$1.472\text{e-}2$	–	$4.564\text{e-}3$	–	$1.308\text{e-}1$	–	$1.356\text{e-}2$	–	1.315e+2
	20×20	$3.815\text{e-}3$	1.95	$1.102\text{e-}3$	2.05	$6.533\text{e-}2$	1.00	$4.561\text{e-}3$	1.57	4.659e+2
	40×40	$9.712\text{e-}4$	1.97	$2.704\text{e-}4$	2.03	$3.263\text{e-}2$	1.00	$1.562\text{e-}3$	1.54	1.228e+3
	80×80	$2.451\text{e-}4$	1.99	$6.724\text{e-}5$	2.01	$1.631\text{e-}2$	1.00	$5.341\text{e-}4$	1.55	3.579e+3
10^{-8}	10×10	$1.472\text{e-}2$	–	$4.562\text{e-}3$	–	$1.308\text{e-}1$	–	$1.358\text{e-}2$	–	1.505e+2
	20×20	$3.814\text{e-}3$	1.95	$1.101\text{e-}3$	2.05	$6.533\text{e-}2$	1.00	$4.574\text{e-}3$	1.57	7.309e+2
	40×40	$9.709\text{e-}4$	1.97	$2.698\text{e-}4$	2.03	$3.263\text{e-}2$	1.00	$1.575\text{e-}3$	1.54	3.515e+3
	80×80	$2.449\text{e-}4$	1.99	$6.673\text{e-}5$	2.02	$1.631\text{e-}2$	1.00	$5.494\text{e-}4$	1.52	2.265e+4

Table 2

Numerical results of Example 1 using the biquadratic CUFVM on uniform grids.

a	$N_x \times N_y$	$\ e_h\ _\infty$	CR	$\ e_h\ _{L^2}$	CR	$ e_h _{H^1}$	CR	$ e_h _{1,\Omega_h}$	CR	Cond
1	10×10	$3.894\text{e-}5$	–	$9.192\text{e-}6$	–	$5.618\text{e-}4$	–	$2.896\text{e-}5$	–	2.106e+2
	20×20	$6.126\text{e-}6$	2.66	$1.334\text{e-}6$	2.78	$1.405\text{e-}4$	2.00	$7.249\text{e-}6$	2.00	8.446e+2
	40×40	$1.126\text{e-}6$	2.44	$2.378\text{e-}7$	2.49	$3.513\text{e-}5$	2.00	$1.813\text{e-}6$	2.00	3.382e+3
	80×80	$2.328\text{e-}7$	2.28	$5.174\text{e-}8$	2.20	$8.783\text{e-}6$	2.00	$4.532\text{e-}7$	2.00	1.353e+4
10^{-3}	10×10	$8.716\text{e-}5$	–	$1.300\text{e-}5$	–	$8.417\text{e-}4$	–	$3.284\text{e-}5$	–	4.938e+2
	20×20	$1.123\text{e-}5$	2.96	$1.624\text{e-}6$	3.00	$2.104\text{e-}4$	2.00	$4.144\text{e-}6$	2.99	1.262e+3
	40×40	$1.428\text{e-}6$	2.98	$2.030\text{e-}7$	3.00	$5.261\text{e-}5$	2.00	$5.116\text{e-}7$	3.02	3.473e+3
	80×80	$1.808\text{e-}7$	2.98	$2.537\text{e-}8$	3.00	$1.315\text{e-}5$	2.00	$6.373\text{e-}8$	3.00	1.299e+4
10^{-8}	10×10	$8.717\text{e-}5$	–	$1.300\text{e-}5$	–	$8.417\text{e-}4$	–	$3.359\text{e-}5$	–	7.191e+2
	20×20	$1.123\text{e-}5$	2.96	$1.624\text{e-}6$	3.00	$2.104\text{e-}4$	2.00	$4.306\text{e-}6$	2.96	2.746e+3
	40×40	$1.425\text{e-}6$	2.98	$2.030\text{e-}7$	3.00	$5.261\text{e-}5$	2.00	$5.452\text{e-}7$	2.98	1.417e+4
	80×80	$1.795\text{e-}7$	2.99	$2.537\text{e-}8$	3.00	$1.315\text{e-}5$	2.00	$6.858\text{e-}8$	2.99	1.080e+5

All of these indicate numerical stability of the bilinear and biquadratic CUFVMs in some extent. We also would like to note that the condition numbers of the systems produced by the bilinear CUFVM are roughly 4 times smaller than that by the biquadratic CUFVM on the same grids as observed.

According to the analysis in Section 2.3, especially the condition (39), we know that if $2N_x = N_y$ (i.e., $h_x = 2h_y$) then the resulted discrete system forms a M -matrix. Thus we also tested the bilinear and biquadratic CUFVMs on non-uniform grids with sizes $N_x \times N_y = 7 \times 14$, 14×28 , 28×56 , and 56×112 (thus at each grid level they have similar number of grid points as those in the uniform grids). The computational results are reported in Table 3 and Table 4. It is easy to find that in both

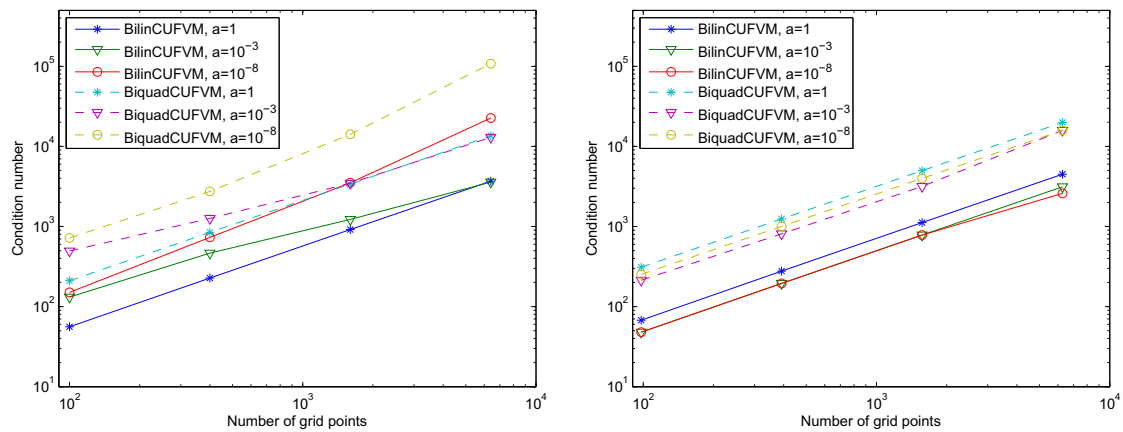


Fig. 7. Plots of condition numbers of the discrete systems produced by the bilinear and biquadratic CUFVMs in Example 1. Left: on the uniform grids; right: on the non-uniform grids.

Table 3

Numerical results of Example 1 using the bilinear CUFVM on non-uniform grids.

a	$N_x \times N_y$	$\ e_h\ _\infty$	CR	$\ e_h\ _{L^2}$	CR	$ e_h _{H^1}$	CR	$ e_h _{1,\Omega_h}$	CR	Cond
1	7×14	1.916e-2	–	7.336e-3	–	1.473e-1	–	9.882e-3	–	6.772e+1
	14×28	4.994e-3	1.94	1.836e-3	2.00	7.364e-2	1.00	2.472e-3	2.00	2.776e+2
	28×56	1.275e-3	1.97	4.592e-4	2.00	3.682e-2	1.00	6.181e-4	2.00	1.120e+3
	56×112	3.221e-4	1.98	1.148e-4	2.00	1.841e-2	1.00	1.545e-4	2.00	4.501e+3
10^{-3}	7×14	1.875e-2	–	6.232e-3	–	1.477e-1	–	1.517e-2	–	4.800e+1
	14×28	4.878e-3	1.94	1.516e-3	2.04	7.376e-2	1.00	4.905e-3	1.63	1.950e+2
	28×56	1.244e-3	1.97	3.734e-4	2.02	3.685e-2	1.00	1.639e-3	1.58	7.829e+2
	56×112	3.144e-4	1.98	9.291e-5	2.00	1.842e-2	1.00	5.517e-4	1.57	3.133e+3
10^{-8}	7×14	1.875e-2	–	6.232e-3	–	1.477e-1	–	1.518e-2	–	4.800e+1
	14×28	4.878e-3	1.94	1.515e-3	2.04	7.376e-2	1.00	4.918e-3	1.63	1.950e+2
	28×56	1.244e-3	1.97	3.728e-4	2.02	3.685e-2	1.00	1.652e-3	1.57	7.830e+2
	56×112	3.141e-4	1.99	9.242e-5	2.01	1.842e-2	1.00	5.680e-4	1.54	2.587e+3

Table 4

Numerical results of Example 1 using the biquadratic CUFVM on non-uniform grids.

a	$N_x \times N_y$	$\ e_h\ _\infty$	CR	$\ e_h\ _{L^2}$	CR	$ e_h _{H^1}$	CR	$ e_h _{1,\Omega_h}$	CR	Cond
1	7×14	1.424e-4	–	3.053e-5	–	1.253e-3	–	1.436e-4	–	3.109e+2
	14×28	2.362e-5	2.59	5.008e-6	2.61	3.135e-4	2.00	3.608e-5	1.99	1.245e+3
	28×56	4.667e-6	2.34	1.024e-6	2.29	7.838e-5	2.00	9.034e-6	2.00	4.979e+3
	56×112	1.019e-6	2.20	2.396e-7	2.10	1.959e-5	2.00	2.259e-6	2.00	1.991e+4
10^{-3}	7×14	1.412e-4	–	2.696e-5	–	1.251e-3	–	2.813e-5	–	2.136e+2
	14×28	1.830e-5	2.95	3.372e-6	3.00	3.129e-4	2.00	3.432e-6	3.03	8.101e+2
	28×56	2.332e-6	2.97	4.216e-7	3.00	7.825e-5	2.00	4.254e-7	3.01	3.167e+3
	56×112	2.938e-7	2.99	5.270e-8	3.00	1.956e-5	2.00	5.636e-8	2.92	1.587e+4
10^{-8}	7×14	1.412e-4	–	2.696e-5	–	1.251e-3	–	2.907e-5	–	2.523e+2
	14×28	1.830e-5	2.95	3.372e-6	3.00	3.129e-4	2.00	3.620e-6	3.00	1.003e+3
	28×56	2.330e-6	2.97	4.216e-7	3.00	7.825e-5	2.00	4.515e-7	3.00	3.985e+3
	56×112	2.938e-7	2.99	5.270e-8	3.00	1.956e-5	2.00	5.636e-8	3.00	1.587e+4

methods the errors and convergence rates are almost equal to those obtained on the uniform grids for all cases although the condition numbers of the systems have some improvements (roughly 4 to 10 times smaller) for the strongly convection-dominated case $a = 10^{-8}$. On these non-uniform grids the condition numbers of the discrete systems increase almost perfectly by a rate of 4 for all cases along the uniform mesh refinements in both methods, see Fig. 7 (right).

We also report the test results by the UFVM and the OWUCM in Tables 5 and 6 respectively for convection-dominated cases on the uniform grids for comparison. It is known that both the UFVM and the OWUCM result in M -matrix and are quite

Table 5

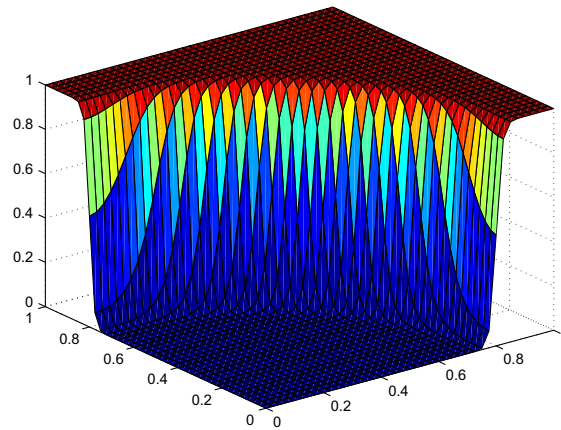
Numerical results of Example 1 using the UFVM on uniform grids.

a	$N_x \times N_y$	$\ e_h\ _\infty$	CR	$\ e_h\ _{L^2}$	CR	$ e_h _{H^1}$	CR	$ e_h _{1,\Omega_h}$	CR	Cond
10^{-3}	10×10	2.020e-1	–	6.460e-2	–	5.815e-1	–	5.592e-1	–	5.218e+1
	20×20	1.179e-1	0.78	3.838e-2	0.75	4.510e-1	0.37	4.433e-1	0.33	2.059e+2
	40×40	6.183e-2	0.93	2.020e-2	0.93	3.257e-1	0.47	3.230e-1	0.46	8.030e+2
	80×80	3.366e-2	0.88	1.154e-2	0.81	2.416e-1	0.43	2.407e-1	0.42	3.196e+3
10^{-8}	10×10	2.274e-1	–	7.994e-2	–	6.282e-1	–	6.052e-1	–	5.287e+1
	20×20	1.282e-1	0.83	4.352e-2	0.88	4.812e-1	0.38	4.734e-1	0.35	2.084e+2
	40×40	6.787e-2	0.92	2.269e-2	0.94	3.538e-1	0.44	3.512e-1	0.43	8.189e+2
	80×80	3.489e-2	0.96	1.158e-2	0.97	2.551e-1	0.47	2.542e-1	0.47	3.239e+3

Table 6

Numerical results of Example 1 using the OWUCM on uniform grids.

a	$N_x \times N_y$	$\ e_h\ _\infty$	CR	$\ e_h\ _{L^2}$	CR	$ e_h _{H^1}$	CR	$ e_h _{1,\Omega_h}$	CR	Cond
10^{-3}	10×10	2.020e-1	–	6.460e-2	–	5.815e-1	–	5.592e-1	–	5.286e+1
	20×20	1.179e-1	0.78	3.838e-2	0.75	4.510e-1	0.37	4.433e-1	0.33	2.083e+2
	40×40	6.183e-2	0.93	2.020e-2	0.93	3.257e-1	0.47	3.230e-1	0.46	8.185e+2
	80×80	2.997e-2	1.04	9.917e-3	1.03	2.210e-1	0.56	2.201e-1	0.55	3.234e+3
10^{-8}	10×10	2.054e-1	–	6.575e-2	–	5.906e-1	–	5.685e-1	–	5.288e+1
	20×20	1.319e-1	0.64	3.971e-2	0.73	4.596e-1	0.36	4.583e-1	0.31	2.098e+2
	40×40	6.620e-2	1.00	2.170e-2	0.87	3.481e-1	0.40	3.454e-1	0.41	8.192e+2
	80×80	3.446e-2	0.94	1.133e-2	0.94	2.530e-1	0.46	2.521e-1	0.46	3.267e+3

**Fig. 8.** Plot of the exact solution u defined in (55) in Example 2.**Table 7**

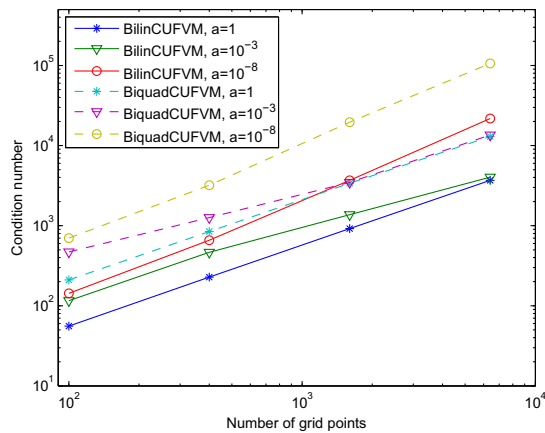
Numerical results of Example 2 using the bilinear CUFVM on uniform grids.

a	$N_x \times N_y$	$\ e_h\ _\infty$	CR	$\ e_h\ _{L^2}$	CR	$ e_h _{H^1}$	CR	$ e_h _{1,\Omega_h}$	CR	Cond
1	10×10	3.711e-1	–	8.639e-2	–	3.426e+0	–	2.811e+0	–	5.583e+1
	20×20	2.138e-1	0.80	3.402e-2	1.34	2.309e+0	0.57	1.333e+0	1.08	2.281e+2
	40×40	5.810e-2	1.88	9.083e-3	1.91	1.262e+0	0.87	5.077e-1	1.39	9.183e+2
	80×80	1.586e-2	1.88	2.287e-3	1.99	6.366e-1	0.99	1.385e-1	1.87	3.684e+3
10^{-3}	10×10	4.935e-1	–	1.187e-1	–	4.685e+0	–	3.042e+0	–	1.157e+2
	20×20	2.092e-1	1.24	4.304e-2	1.46	3.082e+0	0.60	1.736e+0	0.81	4.658e+2
	40×40	5.899e-2	1.83	8.657e-3	2.31	1.308e+0	1.23	4.240e-1	2.03	1.372e+3
	80×80	1.562e-2	1.92	2.103e-3	2.04	6.331e-1	1.05	1.036e-1	2.03	4.035e+3
10^{-8}	10×10	5.410e-1	–	1.238e-1	–	4.866e+0	–	3.125e+0	–	1.429e+2
	20×20	2.317e-1	1.22	4.857e-2	1.35	3.493e+0	0.48	2.002e+0	0.64	6.592e+2
	40×40	5.901e-2	1.97	9.831e-3	2.30	1.525e+0	1.20	5.396e-1	1.89	3.685e+3
	80×80	1.562e-2	1.92	2.092e-3	2.23	6.333e-1	1.27	1.026e-1	2.39	2.175e+4

Table 8

Numerical results of Example 2 using the biquadratic CUFVM on uniform grids.

a	$N_x \times N_y$	$\ e_h\ _\infty$	CR	$\ e_h\ _{L^2}$	CR	$ e_h _{H^1}$	CR	$ e_h _{1,\Omega_h}$	CR	Cond
1	10×10	2.043e-1	–	1.768e-2	–	1.210e+0	–	1.041e+0	–	2.106e+2
	20×20	5.841e-2	1.81	3.130e-3	2.50	4.898e-1	1.30	3.864e-1	1.43	8.446e+2
	40×40	1.045e-2	2.48	6.306e-4	2.31	1.487e-1	1.72	9.376e-2	2.04	3.382e+3
	80×80	1.882e-3	2.47	1.352e-4	2.22	4.053e-2	1.88	2.196e-2	2.09	1.299e+4
10^{-3}	10×10	2.720e-1	–	2.894e-2	–	1.750e+0	–	2.911e+0	–	4.699e+2
	20×20	8.448e-2	1.69	7.101e-3	2.03	6.485e-1	1.43	9.523e-1	1.61	1.262e+3
	40×40	1.315e-2	2.68	8.055e-4	3.14	1.623e-1	2.00	1.677e-1	2.51	3.473e+3
	80×80	1.733e-3	2.94	1.191e-4	2.76	3.940e-2	2.04	9.783e-3	4.09	1.353e+4
10^{-8}	10×10	2.802e-1	–	3.245e-2	–	1.972e+0	–	3.612	–	6.983e+2
	20×20	8.732e-2	1.68	9.584e-3	1.76	8.863e-1	1.15	1.666e+0	1.12	3.201e+3
	40×40	1.372e-2	2.67	1.626e-3	2.56	2.838e-1	1.64	6.030e-1	1.47	1.956e+4
	80×80	1.490e-3	3.20	8.105e-5	4.32	4.001e-2	2.83	3.225e-2	4.22	1.060e+5

**Fig. 9.** Plots of condition numbers of the discrete systems produced by the bilinear and biquadratic CUFVMs on the uniform grids in Example 2.**Table 9**

Numerical results of Example 2 using the UFVM on uniform grids.

a	$N_x \times N_y$	$\ e_h\ _\infty$	CR	$\ e_h\ _{L^2}$	CR	$ e_h _{H^1}$	CR	$ e_h _{1,\Omega_h}$	CR	Cond
10^{-3}	10×10	6.287e-1	–	1.307e-1	–	3.777e+0	–	3.224e+0	–	5.473e+1
	20×20	4.846e-1	0.38	7.989e-2	0.71	3.023e+0	0.32	2.478e+0	0.38	2.207e+2
	40×40	2.790e-1	0.80	4.444e-2	0.85	2.036e+0	0.57	1.756e+0	0.50	8.776e+2
	80×80	1.466e-1	0.93	2.337e-2	0.93	1.159e+0	0.81	9.995e-1	0.81	3.557e+3
10^{-8}	10×10	6.322e-1	–	1.310e-1	–	3.785e+0	–	3.229e+0	–	5.548e+1
	20×20	4.906e-1	0.37	8.021e-2	0.71	3.043e+0	0.31	2.499e+0	0.37	2.234e+2
	40×40	2.830e-1	0.79	4.473e-2	0.84	2.062e+0	0.56	1.782e+0	0.49	8.965e+2
	80×80	1.480e-1	0.93	2.355e-2	0.93	1.176e+0	0.81	1.017e+0	0.81	3.609e+3

Table 10

Numerical results of Example 3 using the bilinear CUFVM on uniform grids.

a	$N_x \times N_y$	$\ e_h\ _\infty$	CR	$\ e_h\ _{L^2}$	CR	$ e_h _{H^1}$	CR	$ e_h _{1,\Omega_h}$	CR	Cond
1	10×10	1.031e-2	–	4.360e-3	–	1.004e-1	–	5.253e-3	–	7.528e+1
	20×20	2.645e-3	1.96	1.093e-3	2.00	5.020e-2	1.00	1.321e-3	2.00	2.967e+2
	40×40	6.690e-4	1.98	2.734e-4	2.00	2.510e-2	1.00	3.311e-4	2.00	1.176e+3
	80×80	1.682e-4	1.99	6.849e-5	2.00	1.255e-2	1.00	8.287e-5	2.00	4.687e+3
10^{-3}	10×10	1.377e-2	–	4.394e-3	–	1.306e-1	–	1.043e-2	–	1.208e+2
	20×20	3.564e-3	1.95	1.078e-3	2.03	6.526e-2	1.00	3.270e-3	1.67	4.207e+2
	40×40	9.073e-4	1.97	2.673e-4	2.01	3.262e-2	1.00	1.071e-3	1.61	1.264e+3
	80×80	2.293e-4	1.98	6.686e-5	2.00	1.630e-2	1.00	3.561e-4	1.59	3.621e+3
10^{-8}	10×10	1.377e-2	–	4.392e-3	–	1.306e-1	–	1.044e-2	–	1.505e+2
	20×20	3.563e-3	1.95	1.077e-3	2.03	6.527e-2	1.00	3.279e-3	1.67	7.560e+2
	40×40	9.064e-4	1.97	2.667e-4	2.01	3.262e-2	1.00	1.186e-3	1.47	3.965e+3
	80×80	2.286e-4	1.99	6.634e-5	2.01	1.630e-2	1.00	3.667e-4	1.69	2.028e+4

Table 11

Numerical results of Example 3 using the biquadratic CUFVM on uniform grids.

a	$N_x \times N_y$	$\ e_h\ _\infty$	CR	$\ e_h\ _{L^2}$	CR	$ e_h _{H^1}$	CR	$ e_h _{1,\Omega_h}$	CR	Cond
1	10×10	1.004e-4	–	1.564e-5	–	8.452e-4	–	1.590e-4	–	2.745e+2
	20×20	2.013e-5	2.32	2.869e-6	2.45	2.113e-4	2.00	4.012e-5	1.99	1.084e+3
	40×40	4.464e-6	2.17	6.357e-7	2.17	5.285e-5	2.00	1.005e-5	2.00	4.307e+3
	80×80	1.047e-6	2.09	1.534e-7	2.05	1.321e-5	2.00	2.516e-6	2.00	1.717e+4
10^{-3}	10×10	8.768e-5	–	1.300e-5	–	8.417e-4	–	3.335e-5	–	4.751e+2
	20×20	1.127e-5	2.96	1.624e-6	3.00	2.104e-4	2.00	4.180e-6	3.00	1.282e+3
	40×40	1.431e-6	2.98	2.030e-7	3.00	5.261e-5	2.00	5.124e-7	3.03	3.514e+3
	80×80	1.810e-7	2.98	2.537e-8	3.00	1.315e-5	2.00	6.230e-8	3.04	1.307e+4
10^{-8}	10×10	8.767e-5	–	1.300e-5	–	8.417e-4	–	3.403e-5	–	8.693e+2
	20×20	1.126e-5	2.96	1.624e-6	3.00	2.104e-4	2.00	4.336e-6	2.97	3.230e+3
	40×40	1.427e-6	2.98	2.030e-7	3.00	5.262e-5	2.00	5.471e-7	2.99	1.493e+4
	80×80	1.796e-7	3.00	2.537e-8	3.00	1.315e-5	2.00	6.870e-8	2.99	1.044e+5

Table 12

Numerical results of Example 3 using the bilinear CUFVM on non-uniform grids.

a	$N_x \times N_y$	$\ e_h\ _\infty$	CR	$\ e_h\ _{L^2}$	CR	$ e_h _{H^1}$	CR	$ e_h _{1,\Omega_h}$	CR	Cond
1	7×14	1.909e-2	–	7.264e-3	–	1.473e-1	–	9.805e-3	–	9.341e+1
	14×28	4.980e-3	1.94	1.819e-3	2.00	7.364e-2	1.00	2.451e-3	2.00	3.662e+2
	28×56	1.272e-3	1.97	4.550e-4	2.00	3.682e-2	1.00	6.130e-4	2.00	1.446e+3
	56×112	3.217e-4	1.98	1.138e-4	2.00	1.841e-2	1.00	1.532e-4	2.00	5.750e+3
10^{-3}	7×14	1.775e-2	–	5.984e-3	–	1.476e-1	–	1.390e-2	–	5.600e+1
	14×28	4.618e-3	1.94	1.482e-3	2.01	7.373e-2	1.00	4.222e-3	1.72	2.100e+2
	28×56	1.179e-3	1.97	3.690e-4	2.01	3.684e-2	1.00	1.345e-3	1.65	8.119e+2
	56×112	3.217e-4	1.87	1.138e-4	1.70	1.841e-2	1.00	4.142e-4	1.69	3.751e+3
10^{-8}	7×14	1.775e-2	–	5.981e-3	–	1.476e-1	–	1.392e-2	–	5.600e+1
	14×28	4.617e-3	1.94	1.481e-3	2.01	7.373e-2	1.00	4.232e-3	1.72	2.100e+2
	28×56	1.178e-3	1.97	3.684e-4	2.01	3.684e-2	1.00	1.357e-3	1.64	8.120e+2
	56×112	2.974e-4	1.99	9.185e-5	2.00	1.841e-2	1.00	4.528e-4	1.58	3.192e+3

Table 13

Numerical results of Example 3 using the biquadratic CUFVM on non-uniform grids.

a	$N_x \times N_y$	$\ e_h\ _\infty$	CR	$\ e_h\ _{L^2}$	CR	$ e_h _{H^1}$	CR	$ e_h _{1,\Omega_h}$	CR	Cond
1	7×14	1.927e-4	–	3.001e-5	–	1.262e-3	–	3.237e-4	–	4.103e+2
	14×28	4.559e-5	2.08	5.513e-6	2.44	3.156e-4	2.00	8.181e-5	1.98	1.607e+3
	28×56	1.122e-5	2.02	1.223e-6	2.17	7.893e-5	2.00	2.052e-5	1.99	6.357e+3
	56×112	2.794e-6	2.01	2.954e-7	2.05	1.973e-5	2.00	5.135e-6	2.00	2.530e+4
10^{-3}	7×14	1.412e-4	–	2.695e-5	–	1.251e-3	–	2.897e-5	–	2.307e+2
	14×28	1.830e-5	2.95	3.372e-6	3.00	3.129e-4	2.00	3.494e-6	3.05	8.397e+2
	28×56	2.332e-6	2.97	4.216e-7	3.00	7.825e-5	2.00	4.306e-7	3.02	3.225e+3
	56×112	2.950e-7	2.98	5.270e-8	3.00	1.956e-5	2.00	5.864e-8	2.88	1.269e+4
10^{-8}	7×14	1.412e-4	–	2.695e-5	–	1.251e-3	–	2.984e-5	–	2.712e+2
	14×28	1.830e-5	2.95	3.372e-6	3.00	3.129e-4	2.00	3.670e-6	3.02	1.036e+3
	28×56	2.330e-6	2.97	4.216e-7	3.00	7.825e-5	2.00	4.547e-7	3.01	4.050e+3
	56×112	2.938e-7	2.99	5.270e-8	3.00	1.956e-5	2.00	5.657e-8	3.00	1.600e+4

stable. As expected the discrete systems produced by the UFVM and OWUCM always have the smallest condition numbers among these four methods, but not more than 8 smaller than those by the bilinear CUFVM even for the strongly convection dominated case $a = 10^{-8}$ in our experiments. On the other hand in both the two methods we only obtain roughly first-order convergence for $\|e_h\|_\infty$ and $\|e_h\|_{L^2}$, and half-order convergence for $|e_h|_{H^1}$ and $|e_h|_{1,\Omega_h}$. In addition $|e_h|_{1,\Omega_h}$ almost has no improvement at all in values over $|e_h|_{H^1}$. It is also easy to find the errors produced by both the UFVM and the OWUCM on the 80×80 grid are even several times larger than those by the bilinear and biquadratic CUFVMs on the 10×10 grid.

Example 2. We test the model problem with pure Dirichlet boundary condition but a non-constant convection coefficient $\vec{b}(\mathbf{x}) = (2 - x^2y, 1 + xy^2)$ is used in this example. This convection coefficient can be regarded as a combination of the constant convection coefficient $(2, 1)$ used in Example 1 and some perturbations. Note that it still holds $\nabla \cdot \vec{b} = 0$. Again the diffusion coefficient is set to be $a(\mathbf{x}) = 1, 10^{-3}, 10^{-8}$, respectively, and the reaction coefficient $r(\mathbf{x}) = 1$. The exact solution is chosen to be

Table 14
Numerical results of Example 3 using the UFVM on uniform grids.

a	$N_x \times N_y$	$\ e_h\ _\infty$	CR	$\ e_h\ _{L^2}$	CR	$ e_h _{H^1}$	CR	$ e_h _{1,\Omega_h}$	CR	Cond
10^{-3}	10×10	2.242e-1	–	7.859e-2	–	7.758e-1	–	7.513e-1	–	5.758e+1
	20×20	1.248e-1	0.85	4.312e-2	0.87	5.877e-1	0.40	5.794e-1	0.37	2.165e+2
	40×40	6.492e-2	0.94	2.255e-2	0.94	4.257e-1	0.47	4.228e-1	0.45	8.240e+2
	80×80	3.365e-2	0.95	1.152e-2	0.97	2.993e-1	0.51	2.983e-1	0.50	3.239e+3
10^{-8}	10×10	2.274e-1	–	7.887e-2	–	7.808e-1	–	7.563e-1	–	5.833e+1
	20×20	1.282e-1	0.83	4.328e-2	0.87	5.952e-1	0.39	5.868e-1	0.37	2.192e+2
	40×40	6.788e-2	0.92	2.263e-2	0.94	4.364e-1	0.45	4.335e-1	0.44	8.399e+2
	80×80	3.490e-2	0.96	1.157e-2	0.97	3.141e-1	0.47	3.132e-1	0.47	3.280e+3

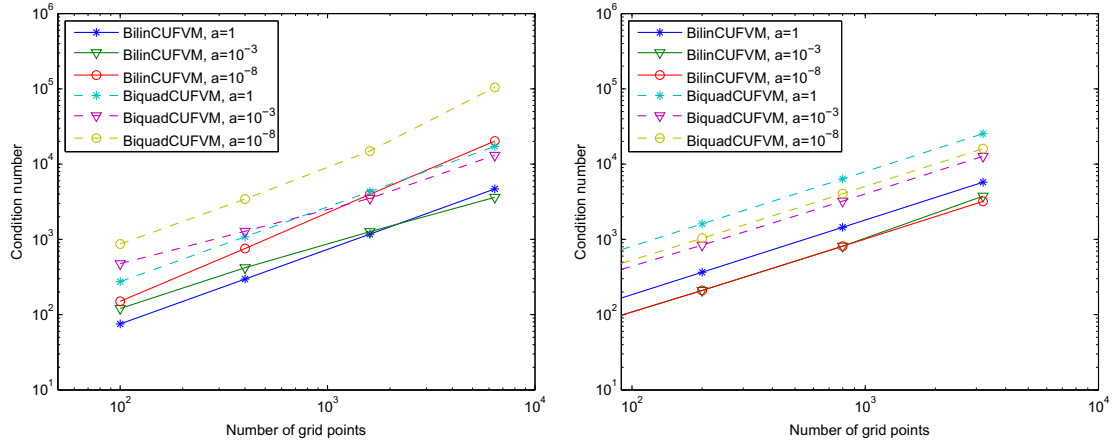


Fig. 10. Plots of condition numbers of the discrete systems produced by the bilinear and biquadratic CUFVMs in Example 3. Left: on the uniform grids; right: on the non-uniform grids.

Table 15
Numerical results of Example 4 using the bilinear CUFVM on uniform grids.

a	N	$\ e_h\ _\infty$	CR	$\ e_h\ _{L^2}$	CR	$ e_h _{H^1}$	CR	$ e_h _{1,\Omega_h}$	CR	Cond
1	10×10	4.354e-1	–	9.808e-2	–	3.450e+0	–	2.786e+0	–	1.379e+2
	20×20	2.431e-1	0.84	3.701e-2	1.41	2.315e+0	0.58	1.326e+0	1.07	5.286e+2
	40×40	5.810e-2	2.06	9.067e-3	2.03	1.263e+0	0.87	5.075e-1	1.39	2.076e+3
	80×80	1.587e-2	1.87	2.274e-3	2.00	6.367e-1	0.99	1.381e-1	1.88	8.247e+3
10^{-3}	10×10	5.383e-1	–	1.273e-1	–	4.988e+0	–	3.143e+0	–	1.056e+3
	20×20	2.934e-1	0.88	4.717e-2	1.43	3.505e+0	0.51	1.848e+0	0.77	1.596e+3
	40×40	1.061e-1	1.47	9.051e-3	2.38	1.368e+0	1.36	4.656e-1	1.99	2.231e+3
	80×80	2.293e-2	2.21	2.142e-3	2.08	6.354e-1	1.11	1.141e-1	2.03	4.095e+3
10^{-8}	10×10	5.917e-1	–	1.342e-1	–	5.247e+0	–	3.246e+0	–	1.785e+3
	20×20	4.244e-1	0.48	5.717e-2	1.23	4.405e+0	0.25	2.216e+0	0.55	1.305e+4
	40×40	1.305e-1	1.70	1.300e-2	2.14	2.180e+0	1.01	6.859e-1	1.69	9.914e+4
	80×80	2.938e-2	2.15	2.267e-3	2.52	6.612e-1	1.72	1.597e-1	2.10	7.713e+5

$$u(\mathbf{x}) = \frac{1}{1 + e^{-100(\sqrt{x^2+y^2}-0.8)}}, \quad (55)$$

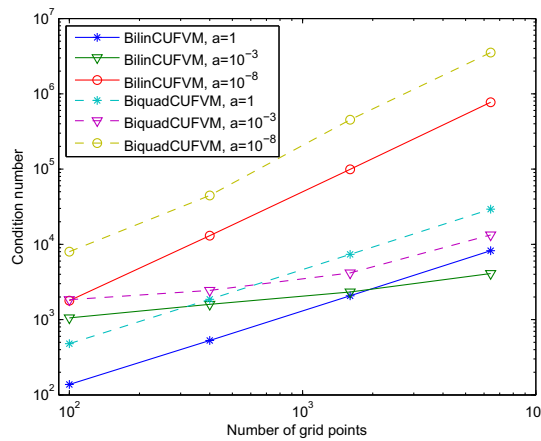
which is smooth but has a sharp change in values across the circle $x^2 + y^2 = 0.8^2$ as shown in Fig. 8. The source function $f(\mathbf{x})$ are again determined by using u and the left-hand side of the Eq. (1) accordingly.

We tested both the linear and biquadratic CUFVMs only on the uniform grids in this example since it is not clear to us what grids will definitely give discretized systems with the M -matrix property. The computational results reported in Tables 7 and 8. It is easy to find that in this example convergence rates of both methods are somewhat smaller than the corresponding optimal ones for all cases when the grid sizes are not small enough and approach closer and closer to the optimal ones along the grid refinements. This phenomenon can be explained by the fact that it is hard to catch the sharp variations of the exact solution u defined in (55) when the grids are not fine enough. Condition numbers of the discrete systems by the two methods (also see Fig. 9) are similar or slightly larger to those with the same grids and diffusion coefficients in Example 1

Table 16

Numerical results of Example 4 using the biquadratic CUFVM on uniform grids.

a	N	$\ e_h\ _\infty$	CR	$\ e_h\ _{L^2}$	CR	$ e_h _{H^1}$	CR	$ e_h _{1,\Omega_h}$	CR	Cond
1	10×10	2.496e-1	–	2.597e-2	–	1.216e+0	–	1.054e+0	–	4.789e+2
	20×20	5.808e-2	2.10	3.139e-3	3.04	4.900e-1	1.31	3.868e-1	1.45	1.862e+3
	40×40	1.054e-2	2.46	6.388e-4	2.30	1.489e-1	1.72	9.431e-2	2.04	7.371e+3
	80×80	1.915e-3	2.46	1.371e-4	2.22	4.055e-2	1.88	2.212e-2	2.09	2.940e+4
10^{-3}	10×10	2.752e-1	–	2.901e-2	–	1.841e+0	–	3.334e+0	–	1.840e+3
	20×20	8.604e-2	1.68	6.947e-3	2.06	6.595e-1	1.48	1.022e+0	1.71	2.446e+3
	40×40	1.391e-2	2.63	8.188e-4	3.08	1.663e-1	1.99	1.688e-1	2.50	4.171e+3
	80×80	1.555e-3	3.16	7.827e-5	3.38	3.947e-2	2.27	1.107e-2	3.53	1.334e+4
10^{-8}	10×10	2.918e-1	–	3.339e-2	–	2.209e+0	–	4.372e+0	–	7.997e+3
	20×20	9.998e-2	1.55	1.010e-2	1.73	1.045e+0	1.08	2.158e+0	1.02	4.365e+4
	40×40	1.519e-2	2.72	1.942e-3	2.38	3.459e-1	1.60	6.197e-1	1.80	4.518e+5
	80×80	2.070e-3	2.88	1.005e-4	4.27	5.475e-2	2.66	9.444e-2	2.71	3.532e+6

**Fig. 11.** Plots of condition numbers of the discrete systems produced by the bilinear and biquadratic CUFVMs on the uniform grids in Example 4.**Table 17**

Numerical results of Example 4 using the UFVM on uniform grids.

a	N	$\ e_h\ _\infty$	CR	$\ e_h\ _{L^2}$	CR	$ e_h _{H^1}$	CR	$ e_h _{1,\Omega_h}$	CR	Cond
10^{-3}	10×10	6.289e-1	–	1.342e-1	–	3.801e+0	–	3.251e+0	–	6.783e+1
	20×20	4.849e-1	0.38	8.134e-2	0.72	3.044e+0	0.32	2.502e+0	0.38	2.451e+2
	40×40	2.792e-1	0.80	4.493e-2	0.86	2.043e+0	0.58	1.766e+0	0.50	9.179e+2
	80×80	1.467e-1	0.93	2.351e-2	0.93	1.158e+0	0.82	9.989e-1	0.82	3.614e+3
10^{-8}	10×10	6.323e-1	–	1.344e-1	–	3.809e+0	–	3.256e+0	–	6.816e+1
	20×20	4.908e-1	0.37	8.165e-2	0.72	3.064e+0	0.32	2.523e+0	0.37	2.475e+2
	40×40	2.831e-1	0.79	4.521e-2	0.85	2.070e+0	0.57	1.793e+0	0.49	9.356e+2
	80×80	1.481e-1	0.93	2.368e-2	0.93	1.175e+0	0.82	1.016e+0	0.82	3.663e+3

although a nonconstant convection coefficient is used in this example. For comparison purpose, we also report the results by the UFVM for this example in Table 9 for convection-dominated cases and again it gave us much worse errors and convergence rates than the CUFVMs.

Example 3. We now test the model problem (1) with mixed boundary conditions. Set $a(\mathbf{x}) = 1, 10^{-3}, 10^{-8}$, respectively, and $r(\mathbf{x}) \equiv 0$. We again use a constant convection coefficient $\tilde{\mathbf{b}}(\mathbf{x}) = (2, 1)$, and choose the exact solution u to be the same function defined in (54) as in Example 1. We set the in-flow boundary $\Gamma_N^{\text{in}} = \emptyset$, the out-flow boundary $\Gamma_N^{\text{out}} = \{x = 1\}$, and the Dirichlet boundary $\Gamma_D = \partial\Omega \setminus \Gamma_N^{\text{out}}$. The boundary condition functions $g(\mathbf{x})$ and $h_{\text{out}}(\mathbf{x})$ are then computed from u and a . The source function $f(\mathbf{x})$ is again determined by u and the model Eq. (1) accordingly.

The computational results by the bilinear and biquadratic CUFVMs on the uniform grids for this example are reported in Tables 10 and 11 and those on the non-uniform grids in Tables 12 and 13. The results by the UFVM for convection-dominated cases on the uniform grids are presented in Table 14. Although an extra outflow boundary condition is imposed (compared with Example 1), but for both methods and all cases we still obtained almost same numerical results as those in Example 1 in $\|e_h\|_\infty$, $\|e_h\|_{L^2}$, and $|e_h|_{1,\Omega_h}$ and their convergence rates and condition numbers (see Fig. 10).

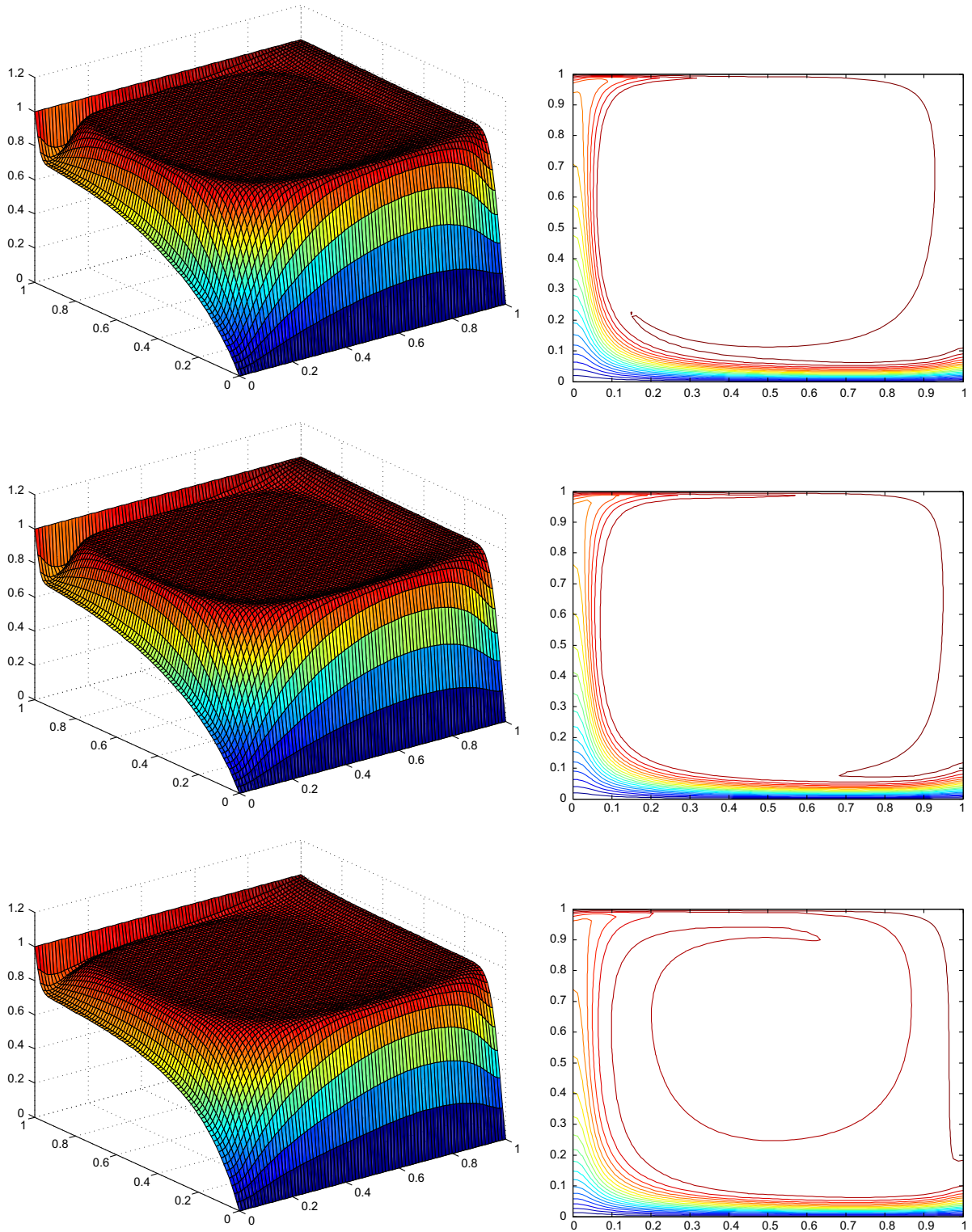


Fig. 12. Plots of the approximate solutions (left) and their corresponding contours (right) for $a = 10^{-4}$ solved on a 80×80 grid in Example 5. From top to bottom: the bilinear CUFVM, the biquadratic CUFVM and the UFVM.

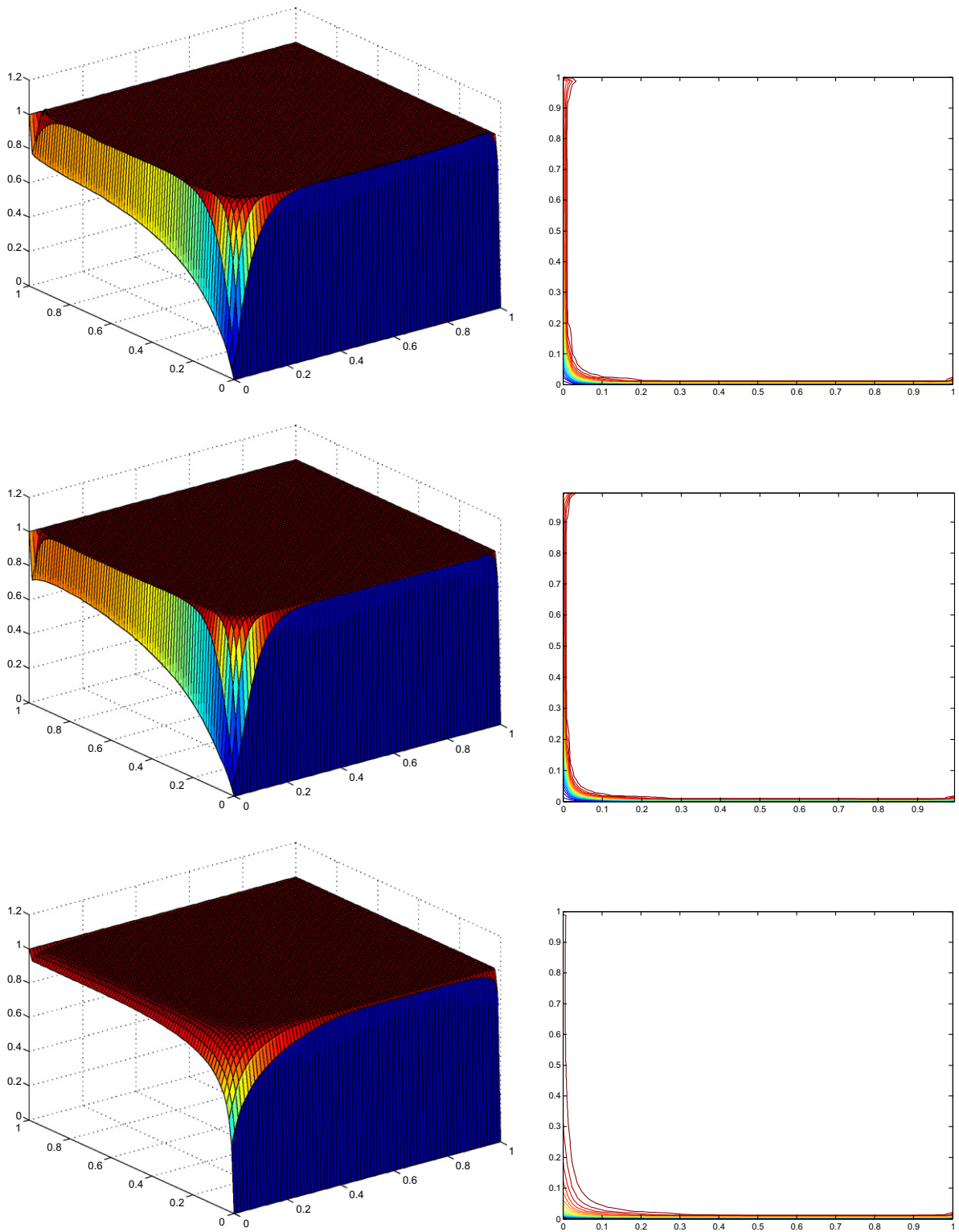


Fig. 13. Plots of the approximate solution (left) and their corresponding contours (right) for $a = 10^{-7}$ solved on a 80×80 grid in Example 5. From top to bottom: the bilinear CUFVM, the biquadratic CUFVM and the UFVM.

Example 4. This example still tests the model problem (1) with mixed boundary conditions, but a non-constant convection coefficient $\vec{b}(\mathbf{x}) = (2 - x^2y, 1 + xy^2)$ is used as in Example 2. We again set $a(\mathbf{x}) = 1, 10^{-3}, 10^{-8}$, respectively, and $r(\mathbf{x}) = 0$. We also choose the exact solution u to be the same one as defined in (55) in Example 2, which has a sharp change around the circle $x^2 + y^2 = 0.8^2$. For the mixed boundaries, we set $\Gamma_N^{in} = \{y = 0\}$, $\Gamma_N^{out} = \{x = 1\}$, and $\Gamma_D = \partial\Omega \setminus (\Gamma_N^{in} \cup \Gamma_N^{out})$. The boundary condition related functions g, h_{in} and h_{out} is then computed from u, a and $\vec{b}$. The source function $f(\mathbf{x})$ is again determined by u and the model Eq. (1) accordingly.

As Example 2 we tested the linear and biquadratic CUFVMs only on the uniform grids. The computational results by the bilinear CUFVM are presented in Table 15 and the results by the biquadratic CUFVM in Table 16. Plot of condition numbers of the discrete system by both methods is given in Fig. 11. For comparison the results by the UFVM for this example are listed in Table 17 for convection-dominated cases. We observe in this example similar behaviors as that in Example 2 in convergence rates (i.e., the convergence rates of both methods are again somewhat smaller than the corresponding optimal ones for all cases when the grid sizes are not small enough but approach closer and closer to the optimal ones along the grid refinements), but the condition numbers on the finest grid 80×80 are around 30 times larger than those in Example 2 correspondingly for the strongly convection-dominated case $a = 10^{-8}$. Note in this example we have both in-flow and out-flow boundaries and that did cause some effects on the conditioning of the discretized systems.

Example 5. This example tests a temperature convection model in a driven cavity domain. The convection coefficient in the model problem (1) is given by

$$\vec{b}(\mathbf{x}) = (16x^2(1-x)^2(3y^2-2y), 16(2x-6x^2+4x^3)y^2(1-y)).$$

Clearly $\nabla \cdot \vec{b}(\mathbf{x}) = 0$ and its corresponding stream function is $\psi(\mathbf{x}) = 16x^2(1-x)^2y^2(1-y)$. The reaction coefficient is taken to be $r(\mathbf{x}) = 0$ and the source function $f(\mathbf{x}) = 0$. We test numerical robustness of the proposed CUFVM methods for convection-dominated problems by setting $a(\mathbf{x}) = 10^{-4}$ and 10^{-7} respectively in this example. The mixed boundary conditions are given by: $u = 1$ on $\{y = 1\}$, $u = 0$ on $\{y = 0\}$, $(-a\nabla u + \vec{b}u) \cdot \vec{n} = 0$ on $\Gamma_N^{in} = \{x = 0\}$, and $-a\nabla u \cdot \vec{n} = 0$ on $\Gamma_N^{out} = \{x = 1\}$. There is no explicit formula for the exact solution, but some boundary layer structures are expected in the solution when a is small.

We solved the above temperature convection problem by using the bilinear CUFVM, the biquadratic CUFVM and the standard UFVM on a 80×80 uniform grid. The approximate solutions and their corresponding contours for the case $a = 10^{-4}$ are presented in Fig. 12 and that for $a = 10^{-7}$ in Fig. 13. We can easily see the boundary layers in both two convection-dominated cases, and the smaller the diffusion coefficient a is, the stronger the boundary layers. Both the bilinear CUFVM and the biquadratic CUFVM are almost diffusion free and catch the sharpness in the boundary regions. However, the UFVM shows relatively large artificial diffusions in the approximate solution and tends to smooth out the sharpness to some extent, which is not hard to tell when one compares the surface and contour plots of the solutions from these three methods in Figs. 12 and 13.

5. Conclusions

In this paper we proposed the bilinear and biquadratic covolume-upwind finite volume methods on rectangular grids for solving linear elliptic partial differential equations with mixed boundary conditions, and demonstrated the stability and optimal convergence of the proposed methods through analysis in some special cases and various numerical examples. These proposed methods can be easily extended to three dimensions. But we also note the methods could not be straightforwardly generalized to produce higher order CUFVMs such as a bicubic one, since uniform partition of the primary grid cell into subcells for constructing higher order methods and the midpoint quadrature rule will fail to catch higher order accuracies.

There are still many interesting and important problems that need further investigations, for example, rigorous and complete analysis of the stability and error estimates of the two proposed CUFVMs, development of adaptive mesh refinement schemes, extensions of such covolume-upwind methods from structured rectangular grids to unstructured quadrilateral grids and/or triangular grids, and generalization of the methods to nonlinear convection–diffusion–reaction problems.

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